

Generative AI and Occupational Entry Barriers: The Labor-Supply Channel of Technological Change*

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Abstract

How will generative AI (GenAI) affect wage inequality? Beyond standard labor-demand effects, GenAI can reshape the wage distribution by changing the *expertise* required to enter occupations, thereby shifting their potential labor supply. We develop a parsimonious general-equilibrium model in which changes in occupational wages reflect a productivity–scarcity race between these labor-supply shifts and productivity gains. We construct measures of GenAI-induced Potential Supply Shifts (PSS)—the change in the share of workers qualified for each occupation—and Potential Productivity Gains (PPG). In our quantitative exercises, the two forces operate in opposite directions: productivity gains widen wage inequality, whereas lower entry barriers compress it. Which force dominates depends on the elasticity of substitution across occupational outputs: inequality falls when this elasticity is low and rises when it is high.

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1 Introduction

The rapid diffusion of generative AI (GenAI) tools since late 2022 and the accelerating pace of model improvement have raised a central question for policy and labor economics: how will GenAI reshape the wage distribution? Much of the existing discussion approaches this question from the labor-demand side, focusing on how GenAI raises productivity, automates subsets of tasks, and creates new ones (e.g., [Eloundou et al., 2024](#)). This emphasis continues a long tradition in labor economics of studying how technology reshapes relative labor demand, from the race between education and technology ([Tinbergen, 1974](#); [Katz and Murphy, 1992](#); [Goldin and Katz, 2008](#)) to task-based models in which technology substitutes for labor in some tasks and complements it in others ([Acemoglu and Autor, 2011](#); [Acemoglu and Restrepo, 2018](#)).

In this paper, we study how GenAI affects the wage distribution across occupations. We argue that, in addition to its demand-side effects, GenAI can reshape occupational outcomes through a labor-supply channel, à la [Autor and Thompson \(2025\)](#), by changing expertise-based entry barriers. We treat occupational expertise as the specialized knowledge, training time, or credentials required to perform an occupation’s tasks at a professional level. GenAI can change these requirements in two ways: it can remove some tasks from the workers’ task bundle, and it can assist workers with the tasks that remain human-performed—for example by supplying specialized knowledge or checking a worker’s output—allowing workers with less training to perform them. In doing so, GenAI can shift the pool of workers qualified to enter an occupation.¹ Where requirements fall, the resulting expansion of potential labor supply increases occupational contestability and can put downward pressure on wage premia. GenAI’s overall distributional impact therefore depends on the interaction between demand-side forces, such as productivity gains and automation, and supply-side forces operating through occupational entry barriers.

We begin in Section 3 by developing a parsimonious general-equilibrium model of occupations and expertise, inspired by [Autor and Thompson \(2025\)](#). Workers differ in a one-dimensional index of expertise, and occupations differ in the expertise they re-

¹[Autor \(2024\)](#) suggests that AI can “enable a larger set of workers possessing complementary knowledge to perform some of the higher-stakes decision-making tasks currently arrogated to elite experts like doctors, lawyers, coders and educators.”

quire. Workers can enter any occupation whose requirement they meet, so a worker who qualifies for a high-requirement occupation also qualifies for all lower-requirement occupations (we later relax this one-dimensionality assumption in the dynamic extension). Among the occupations available to them, workers choose based on wages and idiosyncratic preferences, and wages adjust to clear each occupational labor market.

GenAI affects wages through two channels. First, it raises labor productivity, shifting labor demand outward and raising wages most where productivity gains are largest. Second, it changes the expertise occupations require. Where requirements fall, more workers qualify, expanding potential labor supply and eroding scarcity premia. We derive an intuitive first-order approximation that makes this *race between productivity and scarcity* transparent. In partial equilibrium, an occupation’s wage change equals a demand term proportional to its *Potential Productivity Gains* (PPG), the share of labor time that GenAI can save, minus a supply term proportional to its *Potential Supply Shift* (PSS), the change in the share of the workforce qualified to enter the occupation. The elasticity of substitution across occupations governs the relative weight of the two terms.^{2,3}

PSS is the paper’s central object. Each occupation’s expertise requirement aggregates the requirements of its tasks, and GenAI changes this requirement through two margins. On the *extensive margin*, GenAI automates some tasks and removes them from the bundle performed by workers; this lowers the requirement when the automated tasks are relatively difficult and raises it when they are relatively easy. On the *intensive margin*, tasks that remain in the workers’ bundles become easier because workers have access to a GenAI assistant. PSS translates the resulting change in an occupation’s expertise requirement into a change in the share of workers qualified to enter it, given the distribution of expertise in the workforce. Like standard exposure measures (e.g., Eloundou et al., 2024), PSS is constructed from task-level assessments and can be computed without solving the model. But while exposure measures capture which tasks GenAI can automate or augment, i.e., labor demand forces, PSS captures the corresponding supply-side consequence: how much the pool of qualified workers changes.

²The term *productivity–scarcity race* echoes the classical race between education and technology (Tinbergen, 1974; Goldin and Katz, 2008), where technology has been studied primarily through its effect on labor demand. Here, by contrast, we focus on how technology also shifts labor supply.

³Under stronger assumptions, Appendix A.1 derives an exact race condition for the *expertise premium*, the elasticity of wages with respect to occupational expertise requirements. This object is related to, but distinct from, overall wage inequality.

Motivated by the model, Section 4 constructs empirical counterparts to its two channels using O*NET task statements for 19,265 tasks across 923 occupations. We use large language models (LLMs) to rate the expertise required for each task. Tasks that require more education, specialized training, or credentials receive higher scores, while tasks that most workers can perform with minimal instruction receive lower scores. The resulting occupation-level expertise requirements are stable across frontier LLMs and strongly correlated with O*NET measures of required education and with wages, and positively but more weakly with required experience.

We measure the extensive margin by updating the automation exposure measure of Eloundou et al. (2024) to reflect current GenAI capabilities, and the intensive margin by re-rating each task’s expertise under the counterfactual that the worker has access to a capable GenAI assistant.⁴ We provide evidence from the real-world AI usage data that supports the distinction between the two margins. In the Anthropic Economic Index (Appel et al., 2026), tasks with high extensive-margin scores are more often delegated to GenAI and less often associated with learning and assistive uses, while tasks with high intensive-margin scores show the opposite pattern. Finally, to translate these changes into PSS, we approximate the workforce distribution of expertise using the baseline expertise requirements of the occupations in which workers are employed.

On the demand side, we construct PPG by aggregating task-level time savings into an occupation-level productivity index using O*NET task frequency and importance weights. This implementation mirrors the model’s decomposition: PSS captures how GenAI changes occupational entry barriers and potential labor supply, while PPG captures how it changes labor productivity and, through it, labor demand.

Section 4.3 documents how the two measures vary across occupations. PSS generally rises with an occupation’s expertise requirement but falls sharply above roughly the 90th percentile: few workers lie near very high thresholds, so even large reductions in required expertise add few eligible workers. PPG, by contrast, rises steadily with expertise. The supply channel is therefore strongest in high-expertise occupations below the very top, while occupations at the top experience large productivity gains with little offsetting expansion in potential labor supply. PSS also varies widely among occupations with similar

⁴To avoid mechanically anchoring the with-GenAI rating to the baseline rating, we obtain the two assessments from completely independent prompts and API runs.

GenAI exposure levels, so standard exposure measures alone are an incomplete guide to how GenAI changes occupational entry barriers.

Section 5 embeds the empirical measures in the calibrated model to quantify their general-equilibrium implications for occupational wages. To isolate the two channels, we study three counterfactual scenarios: GenAI changes expertise requirements only, raises productivity only, or does both. In the expertise-only scenario, lower entry barriers compress occupational wage premia: they draw workers out of low-wage occupations and into higher-expertise ones, raising wages at the bottom and lowering them in high-wage occupations.⁵ In the productivity-only scenario, productivity gains widen the wage distribution because they are largest in high-wage occupations.

With both channels, the net effect on inequality depends on the productivity–scarcity race, whose outcome turns on the elasticity of substitution across occupations, σ . A higher elasticity means that a productivity gain raises an occupation’s wage by more, and that wages decline more slowly as an occupation employs more workers. In our quantitative results, when substitution is low ($\sigma < 2.7$), lower expertise requirements dominate and inequality falls; when it is high ($\sigma > 2.7$), productivity gains dominate and inequality rises. For example, the variance of log wages across workers falls by 9 percent at an elasticity of 2 but rises by 13 percent at 4 and by 26 percent at 6.⁶

To assess how informative PSS is as a summary measure, we show that it explains about half of the variation across occupations in the general-equilibrium wage effects of the expertise-only scenario. Because PSS, like standard exposure measures, can be computed directly from task-level data without solving the model, it provides a simple summary of GenAI’s supply-side effect.

Section 6 then extends the analysis to a dynamic model in which forward-looking workers face occupation-to-occupation retraining costs, replacing the single expertise threshold of the static model. This captures, for example, that moving from nursing into

⁵This is related to the “ripple effects” in [Acemoglu and Restrepo \(2022\)](#): shocks in one part of the occupational distribution can affect wages elsewhere through worker reallocation. However, the source of reallocation differs: in their model it comes from automation-induced displacement, while here it comes from lower entry barriers in higher-expertise occupations.

⁶The elasticity of substitution across occupational outputs has not been estimated for detailed occupations. [Burstein et al. \(2019\)](#) estimate an elasticity of about 1.8 across 30 broad occupation groups. The roughly 900 detailed occupations in our data are plausibly closer substitutes, implying a higher elasticity. [Autor and Thompson \(2025\)](#) use $\sigma = 4$ in their numerical illustrations.

medicine is far easier than moving from nursing into law, even though the two destinations require similar levels of expertise.⁷ We estimate the retraining distance between every pair of 3-digit SOC occupations using an LLM and validate it against an independent O*NET measure of directed skill gaps. Occupation-specific costs of leaving and entering are calibrated to observed worker flows from a large longitudinal sample of U.S. workers (Schubert et al., 2024). We model GenAI as a reduction in retraining distances that varies across occupation pairs. Consistent with the static supply channel, this lowers the long-run variance of log wages across workers by about 10 percent at our baseline calibration: high-wage cognitive occupations absorb large inflows of workers and their wages fall, while low-wage service occupations shed workers and their wages rise. How fast this happens depends mainly on how quickly GenAI is adopted. To illustrate this, we solve the model under several adoption paths. At one extreme, retraining distances fall immediately to their long-run levels; at the other, they decline gradually over 25 years. The decline in inequality over the first decade differs by roughly a factor of five between these two scenarios.

Our framework builds primarily on Autor and Thompson (2025), who study how computerization changed occupational expertise requirements between 1980 and 2018 through the removal and addition of tasks, and show that wages and employment respond in opposite directions. We extend this framework in several ways. First, we apply it to GenAI—whose distributional consequences are a central policy question—and study its potential effect on occupational expertise requirements and the wage structure. Second, we allow technology to affect expertise through an intensive margin, in which tasks remain human-performed but become easier with AI assistance; in their framework, technology changes expertise requirements only by removing (automating) or adding tasks. Third, our PSS index measures the supply shift itself—how many additional workers become qualified—not only how much requirements fall: the same drop in a requirement can qualify many workers or few, depending on the distribution of worker expertise. Fourth, while their model is used primarily to illustrate the mechanism, we take the framework quantitatively to the data, embedding our measures in a calibrated general-equilibrium model that combines the supply channel with productivity gains and quanti-

⁷The pairwise switching-cost framework nests the hierarchical expertise model as a special case: if switching costs are zero for moves into occupations with equal or lower expertise requirements and infinite otherwise, the retraining matrix reduces to the hierarchical case.

fies GenAI’s net effect on wage inequality. Fifth, in the dynamic extension, we relax their one-dimensional expertise hierarchy by replacing the single threshold with pairwise re-training distances between occupations.

We conclude by noting several limitations. The LLM-based ratings, especially those capturing how GenAI changes task expertise, are hard to validate directly and should be interpreted cautiously. Moreover, we hold the set of tasks and occupations fixed, abstracting from the creation of new tasks and the rebundling of existing ones (e.g., [Acemoglu and Restrepo, 2019](#); [Autor et al., 2024](#); [Garicano et al., 2026](#)). Also, importantly, as our dynamic model shows, the supply-side effects of GenAI may take a long time to materialize: they depend on how quickly workers adopt GenAI and learn to use it effectively, and they require workers to reallocate across occupations. These effects are therefore unlikely to be clearly visible at this early stage of adoption, making the model’s predictions difficult to test with current data.⁸ With these caveats, our main message is that beyond productivity, GenAI can reshape wage inequality by altering occupational entry barriers. The balance between productivity gains and expertise-driven supply expansions will play an important role in determining its distributional effects.

2 Related Literature

The economic analysis of technology-driven wage inequality has a long intellectual history. A tradition dating to [Tinbergen \(1974\)](#) and [Freeman \(1975\)](#), formalized in [Katz and Murphy \(1992\)](#) and synthesized by [Goldin and Katz \(2008\)](#), frames skill premia as the outcome of a *race between education and technology*: skill-biased technology raises the relative demand for skilled labor, while the supply of educated workers determines whether premia widen or compress. In this framework, technology shifts labor demand, while education determines labor supply. Building on [Autor and Thompson \(2025\)](#), we study a different possibility: technology itself—specifically GenAI—can also shift labor supply by changing the expertise required to enter occupations and, with it, the set of workers who qualify.

⁸That said, related evidence is broadly consistent with our framework: [Klein Teeselink and Carey \(2026\)](#) construct an independent LLM-based expertise measure that is highly correlated with ours, and, using job postings across 39 countries and exploiting ChatGPT’s release as a shock, find that expertise-raising automation increases advertised wages while expertise-lowering automation does not.

A related mechanism appears in [Danieli \(2026\)](#), who finds that routine-biased technological change reduced the need for skill within routine occupations, lowering the return to skill and shifting their employment toward lower-skilled workers. Our mechanism is similar in spirit but operates across occupations: GenAI lowers the expertise thresholds that restrict entry into occupations, expanding the pool of qualified workers and compressing occupational wage premia.

Our analysis also builds on the task-based approach to technological change ([Autor et al., 2003](#); [Acemoglu and Autor, 2011](#); [Acemoglu and Restrepo, 2018](#)), in which technology substitutes for labor in some tasks and complements it in others. A large body of recent work applies this approach to AI by constructing occupation-level measures of “exposure” from task content (e.g., [Eloundou et al., 2024](#); [Gmyrek et al., 2025](#); [Hampole et al., 2025](#); [Felten et al., 2023](#)). These measures are typically used to study demand-side channels such as automation and productivity gains, and our PPG index is in this spirit. We introduce a supply-side counterpart: PSS measures how GenAI changes an occupation’s entry barrier and, with it, the pool of workers qualified to enter. Because GenAI can shift both labor demand and labor supply, exposure alone is not sufficient to determine its effect on occupational wages: the effect depends on the race between productivity gains and the expansion of the qualified pool.

A fast-growing empirical literature uses exposure measures to study the labor-market effects of GenAI to date, combining them with administrative employment data, job postings, and online platform outcomes. Several studies find declines in employment and hiring of young workers in highly exposed occupations since the launch of ChatGPT ([Brynjolfsson et al., 2025a](#); [Klein Teeselink, 2025](#); [Massenkoff and McCrory, 2026](#)), especially within adopting firms ([Hosseini and Lichtinger, 2025](#)). Using hundreds of millions of job postings across 39 countries, [Klein Teeselink and Carey \(2026\)](#) find that exposure predicts declines in postings after the release of ChatGPT, while wage responses depend on whether automation raises or lowers expertise requirements. More directly related to our mechanism, [de Souza \(2026\)](#) uses administrative software registry data from Brazil and finds that AI adoption increases employment of low-skilled and young workers by enabling them to perform complex tasks that previously required greater expertise. Similarly, in a randomized experiment, [Cruces et al. \(2026\)](#) find that access to a generative-AI assistant reduces the productivity gap between lower- and higher-education workers on

a business problem-solving task.

Our dynamic extension also relates to the literature on worker reallocation with occupational switching frictions (Artuç et al., 2010; Dix-Carneiro, 2014; Caliendo et al., 2019; Traiberman, 2019). Especially relevant is Traiberman (2019), who studies worker adjustment to import competition in a dynamic occupational choice model with switching costs estimated from observed transitions. Our setting differs in two main ways: the shock reduces retraining requirements rather than shifting labor demand, and we measure pairwise retraining requirements directly using an LLM, while calibrating only occupation-specific adjustment costs to observed worker transitions.

Our analysis is closest to recent task-based approaches that translate GenAI’s task-level capabilities into predictions about wages. Freund and Mann (2025) show that automation transforms jobs by changing the weight of the tasks that remain, so workers with similar exposure can experience very different wage effects depending on their skills. We focus instead on the supply side: how GenAI changes who can enter each occupation, which we summarize with PSS. This includes an intensive margin, absent from their model, through which tasks that remain human-performed become easier.

Althoff and Reichardt (2026) develop and estimate a dynamic model with multidimensional skills, skill accumulation, and occupational choice, in which AI augments, automates, and simplifies tasks. Their simplification channel, which lowers tasks’ skill requirements, is closely related to our intensive margin, and they find that it drives a large decline in wage inequality. Our approach is complementary. We focus on between-occupation wage inequality, which allows us to isolate the two channels in a tractable model, derive closed-form conditions for the race between them, and summarize the supply channel with PSS, which can be computed directly from task data without solving the model. Our quantitative results also differ. In their model, augmentation and automation have small distributional effects, and AI reduces inequality in all their scenarios. In our measures, by contrast, productivity gains are concentrated in high-wage occupations and widen inequality, so the net effect depends on the productivity–scarcity race.⁹ Finally, our dynamic extension uses pairwise retraining distances, so the cost of entering an occupation depends on where the worker starts, whereas they combine a single switching cost

⁹Part of this difference reflects the elasticity of substitution across occupations: they use $\sigma = 1.57$ based on estimates for broad occupation groups, whereas we show that the outcome of the race changes substantially across a range of values.

with differences in workers' skills.

3 Model

This section develops a parsimonious general-equilibrium model of occupational wages and inequality. The economy has a continuum of workers and a finite set of occupations. Each occupation produces a distinct output, and these outputs are combined into a final good. Workers differ in expertise and can work only in occupations whose expertise requirements they meet; among these, they choose based on wages and idiosyncratic preferences. GenAI enters the model as a shock in two ways: it raises each occupation's productivity, and it changes the expertise tasks require, which shifts which workers qualify for each occupation.

The model is designed to isolate, with minimal structure, two channels through which GenAI affects labor markets and that we can measure from task-level data: (i) *productivity gains*, which raise labor productivity and shift labor demand, and (ii) *changes in expertise-based scarcity*, which shift occupational entry barriers and potential labor supply. The framework makes the interaction between these channels transparent and provides the basis for the quantitative counterfactuals in Section 5.

3.1 Occupations and Tasks

There is a finite set of occupations $\mathcal{O}$. Each occupation $o \in \mathcal{O}$ consists of a finite set of tasks T_o . Each task $t \in T_o$ has an expertise requirement $r_{ot} \in \mathbb{R}_+$, representing the minimum training or knowledge needed to perform the task at a professional level, and an importance weight $\varphi_{ot} > 0$, reflecting how critical the task is to the occupation's core function.

Expertise requirement (entry barrier): The occupation-level expertise requirement aggregates task-level requirements via a generalized (power) mean:

$$R_o := \left(\frac{\sum_{t \in T_o} \varphi_{ot} r_{ot}^\rho}{\sum_{t \in T_o} \varphi_{ot}} \right)^{1/\rho}, \quad \rho \geq 1. \quad (1)$$

The parameter ρ controls how much the hardest tasks in an occupation determine its entry barrier. At $\rho = 1$, the barrier is the importance-weighted mean of task requirements. As $\rho \rightarrow \infty$, it converges to $\max_t r_{ot}$. In this limiting case, the entry condition in our model coincides with that of [Autor and Thompson \(2025\)](#), where each worker must perform all of an occupation’s tasks and entry is governed by the most demanding one (the “task-bundling” assumption). [Autor and Thompson \(2025\)](#) show that under task bundling and hierarchical expertise,¹⁰ conditional on entry, a worker’s output does not depend on how far her expertise exceeds the barrier—though her expertise remains valuable because it expands the set of occupations she can enter. The strict max is a natural theoretical benchmark but likely overstates how rigidly one task governs entry: in practice, workers can compensate for weakness in one task with strength in others. The power-mean framework nests both extremes—and any intermediate case—as special cases. Appendix [C.3](#) shows that the paper’s main results remain largely consistent across values of ρ .

3.2 Demand: Occupational Production and Final Demand

Occupational output: Occupation o produces a differentiated intermediate input Y_o by hiring labor in a competitive market:

$$Y_o = A_o L_o, \tag{2}$$

where $A_o > 0$ is occupation-specific labor productivity and L_o is employment in occupation o .

The linear technology in Equation (2) follows from task-level time accounting, in the spirit of [Acemoglu \(2025\)](#). Tasks within an occupation are perfect complements, so every task must be completed in fixed proportions, consistent with the inseparability principle in [Autor and Thompson \(2025\)](#):

$$Y_o = \min_{t \in T_o} \left\{ \frac{\ell_{ot}}{h_{ot}} \right\},$$

where ℓ_{ot} is the labor time allocated to task t and h_{ot} is the labor time the task requires per

¹⁰In [Autor and Thompson \(2025\)](#), hierarchical expertise means that a worker with expertise e_i can complete task t if and only if $e_i \geq t$.

unit of occupational output. Under efficient allocation, $\ell_{ot} = h_{ot}Y_o$ for all t , so $L_o = Y_oH_o$ with $H_o := \sum_{t \in T_o} h_{ot}$, which yields Equation (2) with $A_o = 1/H_o$. Let $\lambda_{ot} := h_{ot}/H_o$ denote task t 's share of baseline labor time. Any technology that reduces h_{ot} for some task lowers H_o and thereby raises A_o .

Final demand: A competitive final-good producer aggregates occupational inputs with a CES technology:

$$Y = \left(\sum_{o \in \mathcal{O}} \vartheta_o^{\frac{1}{\sigma}} Y_o^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1, \quad (3)$$

where $\vartheta_o > 0$ captures demand weights and σ is the elasticity of substitution across occupational outputs.¹¹ The final good is the numeraire.

Under perfect competition, wages equal marginal revenue products. Combining Equations (2)–(3) yields the inverse labor-demand schedule

$$w_o = B_o L_o^{-\frac{1}{\sigma}}, \quad (4)$$

where

$$B_o := \Gamma \vartheta_o^{\frac{1}{\sigma}} A_o^{1-\frac{1}{\sigma}} \quad (5)$$

is an occupation-specific demand–productivity shifter and $\Gamma = Y^{1/\sigma}$ is common to all occupations. Equation (4) highlights the demand-side source of wage dispersion: higher B_o implies higher wages for a given employment level.

3.3 Labor Supply: Feasibility and Occupational Choice

Feasibility: The economy contains a unit mass of workers. Worker i has an expertise level $e_i \in \mathbb{R}_+$ drawn from a continuous distribution F with density f . Expertise determines which occupations a worker can enter: a worker with expertise e qualifies for occupation o if and only if $e \geq R_o$, so her feasible set is $\mathcal{O}(e) := \{o \in \mathcal{O} : e \geq R_o\}$.¹²

¹¹Occupational outputs are gross substitutes ($\sigma > 1$), as in [Autor and Thompson \(2025\)](#), and consistent with estimates of the elasticity of substitution between skill and occupation groups ([Katz and Murphy, 1992](#); [Acemoglu and Autor, 2011](#); [Burststein et al., 2019](#)). Section 5.1 discusses the choice of σ .

¹²The static model treats expertise as a one-dimensional, general index. The dynamic extension in Section 6 relaxes this by replacing the single threshold with a matrix of occupation-to-occupation retraining

Occupational choice: Each worker chooses among her feasible occupations based on wages and idiosyncratic preferences, which we model with a standard discrete-choice framework.¹³ Utility from choosing occupation o is

$$u_{io} = \log w_o + \varepsilon_{io},$$

where ε_{io} are i.i.d. Type-I extreme value with scale $\tau > 0$, and $u_{io} = -\infty$ if $e_i < R_o$.¹⁴ The resulting choice probabilities follow a multinomial logit. In particular, the probability that a worker with expertise e chooses occupation o , given wages w and requirements R , is

$$s_o(e; w, R) = \frac{w_o^{1/\tau}}{\sum_{k \in \mathcal{O}(e)} w_k^{1/\tau}} \mathbf{1}\{e \geq R_o\}. \quad (6)$$

As $\tau \rightarrow 0$, choices concentrate on the highest-wage feasible occupation.

Aggregate labor supply: Employment in occupation $o \in \mathcal{O}$ is

$$L_o^S(w, R) = \int_0^\infty s_o(e; w, R) f(e) de. \quad (7)$$

Define the *eligible pool*

$$S_o(R) := 1 - F(R_o). \quad (8)$$

Employment in o equals the eligible pool times the share of eligible workers who choose o . This separates the scarcity margin (eligibility) from the allocation margin (sorting among feasible options).

3.4 Equilibrium and GenAI Shocks

An equilibrium is a vector of wages $\{w_o\}_{o \in \mathcal{O}}$ and employments $\{L_o\}_{o \in \mathcal{O}}$ such that, for each occupation o ,

$$w_o = B_o L_o^{-1/\sigma} \quad \text{and} \quad L_o = L_o^S(w, R). \quad (9)$$

distances, capturing occupation-specific components of expertise in a flexible way.

¹³This specification—a logit with idiosyncratic preferences over differentiated occupations—is standard in the structural occupational-choice literature (e.g., [Traiberman, 2019](#)) and is the occupational analog of the firm-level wage-setting model of [Card et al. \(2018\)](#).

¹⁴We abstract from non-wage amenities.

GenAI affects outcomes through two channels in the benchmark model. Both operate at the task level and aggregate to occupation-level objects via the structures introduced in Sections 3.1–3.2.

Productivity (demand) channel—Potential Productivity Gains: GenAI reduces the labor time required for some tasks. Let $g_{ot} \in [0, 1]$ denote the fraction of labor time saved on task t in occupation o when a worker uses GenAI. The occupation-level *Potential Productivity Gains* (PPG) equal the work-volume-weighted average of task-level time savings:

$$PPG_o := \sum_{t \in T_o} \lambda_{ot} g_{ot}, \quad (10)$$

where $\lambda_{ot} = h_{ot}/H_o$ are the baseline labor-time shares defined in Section 3.2.¹⁵ Post-GenAI, labor time per unit falls to $H_o^{AI} = (1 - PPG_o) H_o$, which implies a labor-augmenting productivity gain:

$$A_o^{AI} = A_o \cdot \exp(\pi_o), \quad \pi_o = -\log(1 - PPG_o). \quad (11)$$

Under Equations (4)–(5), this shifts labor demand through B_o .

Scarcity (expertise) channel—Potential Supply Shift: GenAI affects expertise-based entry barriers through two margins: which tasks workers perform, and how much expertise the remaining tasks require. On the *extensive margin*, GenAI fully automates some tasks, removing them from the set performed by labor. Let $a_{ot} \in \{0, 1\}$ indicate whether task t is automated. On the *intensive margin*, tasks that remain human-performed become easier: GenAI changes the expertise required from r_{ot} to $\tilde{r}_{ot}$.¹⁶ Combining both margins,

¹⁵The linear aggregation follows from time accounting: every task must be completed, so total time saved is the time-share-weighted sum of task-level savings. The expertise requirement (Equation (1)) is aggregated differently because it concerns *feasibility*, whether a worker can perform the occupation at all: it weights tasks by their importance to the occupation, φ_{ot} , and, for $\rho > 1$, gives extra weight to the hardest tasks. PPG concerns *efficiency*, how much time the occupation’s full task bundle requires, so it weights tasks by their share of labor time, λ_{ot} .

¹⁶This extensive margin resonates with the “expertise exposure” concept of Autor and Thompson (2025), though it focuses on potential shifts induced by GenAI automation rather than on realized changes from past technologies. The intensive margin is closely related to “simplification” in Althoff and Reichardt (2026).

the post-GenAI expertise requirement is

$$\tilde{R}_o := \left(\frac{\sum_{t \in T_o} \varphi_{ot} (1 - a_{ot}) \tilde{r}_{ot}^\rho}{\sum_{t \in T_o} \varphi_{ot} (1 - a_{ot})} \right)^{1/\rho}. \quad (12)$$

The new requirement $\tilde{R}_o$ can be above or below R_o : the intensive margin typically lowers the requirement, but the extensive margin can move it either way, since automating tasks harder than R_o lowers it and automating easier tasks raises it.

The change in the requirement shifts the eligible pool. The resulting *Potential Supply Shift* (PSS), the key supply-side object of the paper, is the change in the share of the workforce that meets the occupation's expertise threshold:

$$\text{PSS}_o := F(R_o) - F(\tilde{R}_o). \quad (13)$$

For a small reduction $\tilde{R}_o = R_o - \Delta R_o$, $\text{PSS}_o \approx f(R_o) \Delta R_o$, so potential inflows are larger when many workers have expertise just below the pre-AI threshold.

3.5 The Productivity–Scarcity Race

Outside options and marginal take-up: Newly eligible workers have feasible alternatives. Holding wages fixed, define the outside-option index just below the threshold:

$$\Omega(R_o^-) := \sum_{k: R_k < R_o} w_k^{1/\tau},$$

which represents wages at the set of all other occupations they can supply their labor to given their requirements. The take-up rate of marginally newly eligible workers at occupation o is

$$s_o^m := \frac{w_o^{1/\tau}}{w_o^{1/\tau} + \Omega(R_o^-)}. \quad (14)$$

At fixed wages, the induced increase in labor supply due to a reduction in expertise barrier is approximately

$$\Delta L_o^S|_w \approx s_o^m \cdot \text{PSS}_o. \quad (15)$$

Local incidence: Let $\varepsilon_o := \partial \log L_o^S / \partial \log w_o$ denote the own-wage elasticity of labor supply (holding requirements fixed). A first-order approximation implies

$$\Delta \log w_o \approx \underbrace{\frac{\sigma - 1}{\sigma + \varepsilon_o} PPG_o}_{\text{productivity}} - \underbrace{\frac{1}{\sigma + \varepsilon_o} s_o^m \frac{PSS_o}{L_o}}_{\text{expertise expansion}}, \quad (16)$$

$$\Delta \log L_o \approx \underbrace{\frac{(\sigma - 1)\varepsilon_o}{\sigma + \varepsilon_o} PPG_o}_{\text{productivity}} + \underbrace{\frac{\sigma}{\sigma + \varepsilon_o} s_o^m \frac{PSS_o}{L_o}}_{\text{expertise expansion}}, \quad (17)$$

where we use the first-order approximation $\pi_o \approx PPG_o$. These expressions make the productivity–scarcity race transparent: PPG_o shifts labor demand and PSS_o shifts labor supply. The productivity term carries a factor $\sigma - 1$ relative to the supply term, so a higher σ tilts the race toward productivity: when occupational outputs are closer substitutes, a productivity gain raises labor demand by more and an inflow of workers depresses the wage by less (see Section 5.3 for more discussion). The expressions are meant for intuition rather than measurement; in the quantitative exercise we solve the full system in Equation (9) numerically, allowing for reallocation across all occupations.

Under stronger distributional assumptions, Appendix A.1 derives an exact race condition, not for wage inequality itself but for the *expertise premium*: how steeply wages rise with occupational expertise requirements. The expertise premium is related to, but distinct from, cross-occupation wage inequality, which also depends on how workers are distributed across occupations.¹⁷ The premium steepens when productivity gains rise with expertise more steeply than the proportional expansion of the eligible pool, with the productivity gradient scaled by $\sigma - 1$. Appendix A.2 estimates both gradients in the data and shows that which one dominates depends on σ .

3.6 Discussion: Automation and Displacement

The benchmark model omits a third channel: displacement, in which firms substitute automation services for labor. In task-based models of automation (Acemoglu and Restrepo, 2022; Acemoglu, 2025), its effect on labor demand depends on the balance between

¹⁷For example, if most workers are concentrated in occupations with similar expertise requirements, the expertise premium can be steep even when overall wage dispersion is low.

a *substitution effect* (automation replaces labor in affected tasks) and a *scale effect* (lower unit costs raise output demand, partially restoring labor demand). Which force dominates depends on objects that are difficult to measure: the effective price of automation services, the within-occupation elasticity of substitution between labor and automation, and the mapping from task-level exposure to these parameters. Appendix A.3 adds this channel through a CES task-level production function that nests the benchmark model, and Appendix A.4 shows that its effect on inequality is highly sensitive to these parameters: depending on the within-occupation elasticity relative to σ , it can strongly widen or compress the wage distribution. By contrast, we measure PPG and PSS directly from task-level data: PPG from exposure indicators and work-volume weights, and PSS from expertise ratings and a proxy for the distribution of worker expertise (Section 4). We therefore treat the productivity–scarcity race as the benchmark and displacement as an extension whose incidence remains an open empirical question.

4 Data, Key Variables, and Descriptive Findings

This section takes the model to the data. The model identifies two occupation-level objects that shape the productivity–scarcity race: the Potential Productivity Gains (PPG_o), which enter the demand side through Equation (11), and the Potential Supply Shift (PSS_o), which enters the supply side through Equation (13). Both are constructed from task-level primitives—time savings g_{ot} , expertise requirements r_{ot} and $\tilde{r}_{ot}$, and automation indicators a_{ot} —while PSS additionally depends on the distribution of expertise in the workforce. We construct their empirical counterparts using O*NET task statements for 19,265 tasks across 923 occupations (O*NET, 2023), together with LLM-based evaluations.¹⁸

Section 4.1 constructs the supply-side objects: the baseline expertise requirement R_o (Equation (1)), the post-GenAI requirement $\tilde{R}_o$ (Equation (12)), and the resulting PSS_o . Section 4.2 constructs the demand-side object PPG_o . We then document how PSS and PPG vary across occupations in Section 4.3 and benchmark PSS against standard exposure

¹⁸Three occupations (Insurance Underwriters, Travel Agents and Telemarketers) have all their tasks classified as automatable, so their post-GenAI requirement is undefined; we drop them. Of the remaining 920 occupations, 889 can be matched to BLS employment, which PSS requires, and 885 of these also to BLS wages, which the model requires. The task- and occupation-level datasets are available at https://github.com/s-mahdihosseini/GenAI_Expertise.

measures in Section 4.4.

4.1 Potential Supply Shift (PSS)

We construct PSS in three steps. First, we measure each occupation’s baseline expertise requirement, R_o . Second, we measure how GenAI changes this requirement through the extensive margin (a_{ot}) and the intensive margin ($\tilde{r}_{ot}$), yielding the post-GenAI requirement $\tilde{R}_o$. Third, we translate the change from R_o to $\tilde{R}_o$ into a change in the share of qualified workers using an empirical approximation of the expertise distribution F .

4.1.1 Step 1: Baseline Expertise of Tasks and Occupations

We follow [Autor and Thompson \(2025\)](#) in defining expertise as a barrier to entry: tasks that require specialized training, credentials, or substantial occupation-specific knowledge score higher, while tasks that most workers can perform with minimal instruction score lower. Using GPT-5.2 and this definition, we rate each O*NET task on a five-point scale from 1 (minimal expertise) to 5 (very high expertise); the prompt is in Appendix B.1.1.

To aggregate to the occupation level, we convert these ordinal categories into training time: the number of months a typical adult needs to learn to perform the task at a professional level. A separate prompt estimates training months for each task (Appendix B.1.2). For each occupation and expertise category, we assign the median training time among the occupation’s tasks in that category, allowing the same category to correspond to different training times across occupations. Table 1 reports the pooled mapping across all occupations.

Let x_{ot}^{noAI} denote the resulting training time for task t in occupation o , the empirical counterpart of the model’s r_{ot} , and let the importance weight φ_{ot} be the O*NET task weight (1 for core and 0.5 for supplemental tasks, following [Eloundou et al. 2024](#)). We set $\rho = 1$, following [Autor and Thompson \(2025\)](#), who measure occupational expertise as the average across tasks (Appendix C.3 reports results for other values of ρ). The baseline

Table 1: Mapping Expertise Categories to Required Training Time

Expertise Category	Description	Training Months
1	No or minimal expertise; generic or basic tasks; learn quickly with little training	0.25
2	Low expertise; short training; limited occupation-specific knowledge	1
3	Moderate expertise; solid occupation-specific knowledge; often requires credentials, apprenticeship, or substantial on-the-job learning	4
4	High expertise; advanced specialized knowledge; significant training, degree, or certification	12
5	Very high expertise; deep specialized expertise; often advanced professional or graduate-level training	60

Notes: The table reports the pooled mapping from the 1–5 expertise scale to required training time: the median of the GPT-5.2 training-month estimates across all tasks in each category. The analysis uses occupation-specific medians.

requirement, the empirical counterpart of R_o , is

$$\chi_o^{noAI} = \frac{\sum_{t \in T_o} \varphi_{ot} x_{ot}^{noAI}}{\sum_{t \in T_o} \varphi_{ot}}. \quad (18)$$

Appendix B.2 validates this measure, showing that it is highly stable across frontier LLMs ($R^2 \geq 0.90$) and strongly correlated with O*NET measures of required education ($R^2 = 0.61$) and with wages, and positively but more weakly with required experience ($R^2 = 0.12$). Appendix Figure B.4a plots the distribution of χ_o^{noAI} . The mean requirement is 9.61 months (SD 12.48), and the median is 5.24 months. Appendix Table B.2 lists occupations at both ends of the distribution: surgeons, physicians, judges, and lawyers at four to six years, and dishwashers, cleaners, and food servers at one to two weeks.

4.1.2 Step 2: The Effect of GenAI on Occupational Expertise Requirements

Extensive margin: The extensive margin captures tasks that GenAI fully automates (a_{ot} in Equation (12)). We identify these tasks by updating the automation exposure measure of Eloundou et al. (2024) to reflect current GenAI capabilities. Using GPT-5.2 and an updated prompt (Appendix B.1.3), we assign each task to one of five categories based on

the share of its components that current GenAI can perform: 1 (0 percent), 2 (0–50 percent), 3 (50–80 percent), 4 (80–100 percent), and 5 (100 percent). The indicator α_{ot} , the empirical counterpart of a_{ot} , equals one for tasks in categories 4–5, that is, tasks that are at least 80 percent automatable. This classifies 4,107 tasks (21.3 percent) as automatable. Occupation-level exposure shares from this measure are strongly correlated with the original measure of Eloundou et al. (2024) (Appendix B.5).

Intensive margin: The intensive margin captures tasks that remain with the worker but become easier, lowering their expertise requirement from r_{ot} to $\tilde{r}_{ot}$. To measure this margin, we re-rate each task’s expertise assuming the worker has access to a GenAI assistant (Appendix B.1.4). The prompt specifies what the assistant cannot do: perform physical actions or in-person inspection, guarantee correctness in high-stakes situations, substitute for human judgment where regulation requires a human, or provide genuine emotional presence or tacit understanding. The with-GenAI rating therefore captures how much easier the task becomes while the judgment, accountability, and physical presence it requires remain with the worker. Applying the occupation-specific mapping from Step 1 to this rating gives the training time with GenAI, x_{ot}^{withAI} —the empirical counterpart of $\tilde{r}_{ot}$.¹⁹ Appendix Figure B.3 compares the ratings with and without GenAI. The share of tasks in category 4 falls from 30.0 percent to 8.4 percent, while the share in category 1 rises from 6.0 to 22.9 percent. In every category except the lowest, more than half of tasks move down at least one category; among category 4 tasks, 79.0 percent move to category 3.

Combining the two margins: Combining both margins gives the empirical counterpart of $\tilde{R}_o$ (Equation (12)) at $\rho = 1$:

$$x_o^{combined} = \frac{\sum_{t \in T_o} \varphi_{ot} (1 - \alpha_{ot}) x_{ot}^{withAI}}{\sum_{t \in T_o} \varphi_{ot} (1 - \alpha_{ot})}. \quad (19)$$

Figure B.4 in Appendix B.3 plots the distributions of occupation-level expertise un-

¹⁹The occupation-specific mapping assigns each category the median training time of the occupation’s own baseline tasks in that category. With GenAI, a task can move to a category in which the occupation has no baseline tasks, so the mapping has no value for it. For these tasks (7.5 percent), we use the pooled value from Table 1, capped at the training time of the occupation’s hardest baseline task, so that no task is assigned a training requirement above the occupation’s highest baseline requirement.

der each scenario. In our measures, most of the decline in expertise requirements comes from the intensive margin. Removing automated tasks alone raises the mean requirement across occupations from 9.6 to 10.0 months and raises it in 62 percent of occupations, because automated tasks tend to be easier than the tasks that remain. Making the remaining tasks easier lowers the mean requirement to 6.0 months, while the two margins together lower it to 6.3 months. A lower automation threshold or greater GenAI capabilities would shift more weight toward the extensive margin.

Validation with real-world AI usage data: If the two margins capture distinct channels, they should be associated with different modes of AI use: automatable tasks should be more likely to be delegated to AI, while tasks for which AI lowers expertise requirements should be more likely to involve learning or guidance. We test this using the Anthropic Economic Index (AEI; Appel et al., 2026), which matches anonymized Claude conversations to O*NET tasks. For each task, it reports the log number of conversations, an average speedup measure (human-only time relative to time with AI), and the shares of conversations classified as “directive” (the user delegates and the AI executes) and “learning” (the user seeks explanations to build skill). We correlate these measures with the automation indicator α_{ot} and the reduction in training months, $d_{ot} = x_{ot}^{noAI} - x_{ot}^{withAI}$ (Figure 1).

The results are consistent with the distinction between the two margins. Both are positively correlated with usage and speedup: tasks that our measures identify as more affected by GenAI are used more frequently with AI and are associated with larger time savings. But the two margins differ sharply in how AI is used. The extensive margin is positively correlated with directive use and negatively correlated with learning use, while the intensive margin shows the opposite pattern. Automatable tasks are therefore more often delegated to AI, while tasks that AI makes easier are more likely to involve workers using AI to build skill while remaining engaged in the task. These correlations are only suggestive, but they provide external evidence that the two margins capture distinct effects of GenAI in practice.

4.1.3 Step 3: Mapping Expertise Changes to Potential Labor Supply Shifts

Measuring the empirical counterparts of $S_o(R) = 1 - F(R_o)$ (Equation (8)) and $PSS_o = F(R_o) - F(\tilde{R}_o)$ (Equation (13)) requires an estimate of the expertise distribution F . We

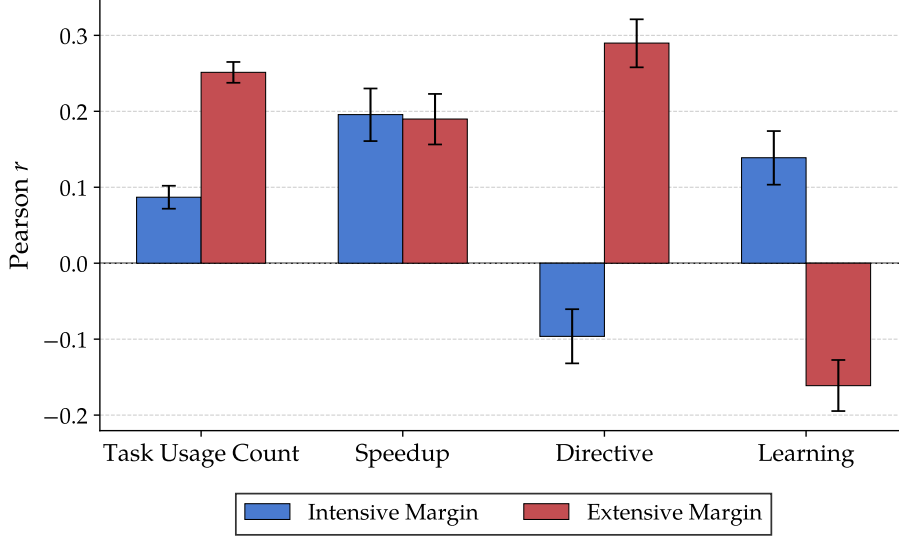


Figure 1: Correlations Between AEI Usage Metrics and Expertise Margins

Notes: The figure reports Pearson correlations between task-level AEI usage metrics and the two margins: the intensive margin (reduction in required training months, blue) and the extensive margin (automation indicator, red). AEI metrics are averaged across the Claude.ai and first-party API platforms. Error bars show 95 percent confidence intervals. Data source: Anthropic Economic Index v4 (Appel et al., 2026).

proxy F with the employment-weighted distribution of baseline occupational requirements X_o^{noAI} and evaluate it at counterfactual requirements by linearly interpolating the empirical CDF.²⁰ Figure 2 plots this distribution. It is right-skewed, with most of its mass at low to moderate expertise levels and a thin upper tail.

The empirical PSS for occupation o is

$$PSS_o = (1 - F(X_o^{combined})) - (1 - F(X_o^{noAI})) = F(X_o^{noAI}) - F(X_o^{combined}). \quad (20)$$

A positive value means that GenAI lowers the entry barrier and expands the share of the workforce qualified for the occupation. The mean PSS is 0.116: for the average occupation, GenAI expands the qualified pool by about 12 percentage points of the workforce (Appendix Figure B.5a). Appendix Table B.3 lists the occupations with the largest and

²⁰This proxy does not invert the model’s latent distribution F : in the model, observed employment reflects both eligibility and idiosyncratic preferences across feasible occupations. Appendix C.4 constructs an alternative distribution from educational attainment in the CPS. The direction of the counterfactual results (Section 5) is unchanged, but the supply shift is smaller.

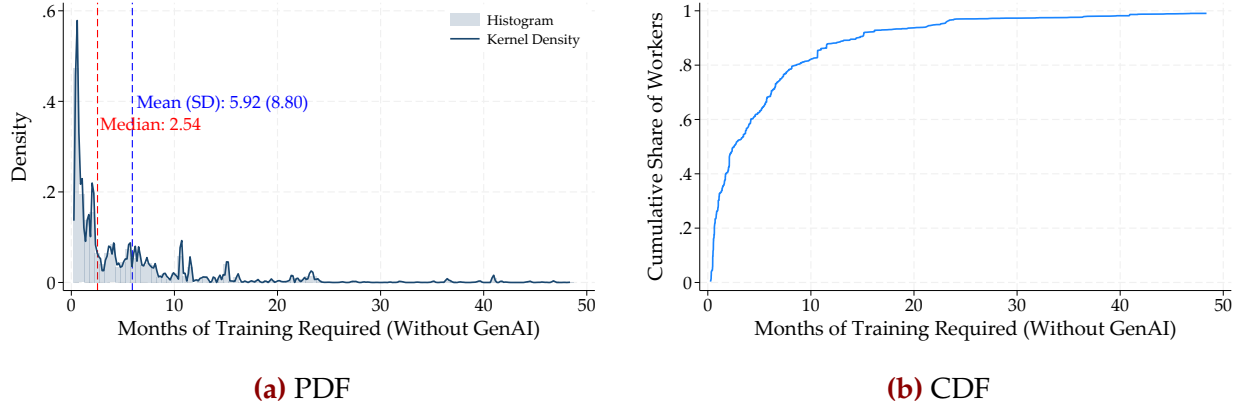


Figure 2: The Distribution of Workers Across Expertise Levels

Notes: Panel (a) shows the density and Panel (b) the cumulative distribution of baseline occupational expertise, weighted by 2024 employment. This distribution is our proxy for the distribution of worker expertise.

smallest PSS. The largest shifts, above 0.3, occur in occupations such as Pharmacy Aides, Credit Analysts, Chemists, and Data Scientists, where the decline in the expertise requirement spans a dense part of the expertise distribution, so many workers become newly qualified.²¹

4.2 The Potential Productivity Gains (PPG) of Occupations

We measure the time savings g_{ot} in Equation (10) with the task-level exposure index β_{ot} of Eloundou et al. (2024): $\beta_{ot} = 1$ for tasks for which an LLM alone cuts the time needed by at least 50 percent, $\beta_{ot} = 0.5$ for tasks for which this requires additional software built on an LLM, and $\beta_{ot} = 0$ otherwise. Because exposure corresponds to time savings of at least 50 percent, we set the savings to 50 percent, the lower bound of this range, so that $g_{ot} = 0.5 \times \beta_{ot}$. We proxy the labor-time shares λ_{ot} with each task's work volume, the product of its O*NET frequency ($Freq_{ot}$) and importance (Imp_{ot}). This gives

$$PPG_o = 0.5 \times \frac{\sum_{t \in T_o} Freq_{ot} \times Imp_{ot} \times \beta_{ot}}{\sum_{t \in T_o} Freq_{ot} \times Imp_{ot}}. \quad (21)$$

²¹At the other end, PSS is negative for 41 occupations (4.8 percent of employment). Their requirement rises with GenAI, either because automation removes easier tasks, raising the average requirement of those that remain, or because some tasks receive a higher expertise rating with GenAI than in the baseline. We keep these values as measured; in the model, barriers can rise as well as fall.

The mean PPG is 0.17 (SD 0.11): for the average occupation, GenAI could reduce total work time by about 17 percent (Appendix Figure B.5b).

4.3 PSS and PPG Across the Occupational Distribution

Figure 3 plots PSS and PPG against each occupation’s baseline expertise percentile, $100 \times F(X_o^{noAI})$ —the share of workers whose expertise lies below the occupation’s requirement. PSS rises with baseline expertise over most of the distribution: GenAI expands the qualified pool more for higher-expertise occupations. Above roughly the 90th percentile, however, PSS falls sharply. This reflects the thin upper tail of the expertise distribution: when an occupation’s requirement lies far in the tail, there is little worker mass near the threshold, so even a large reduction in the requirement adds few qualified workers. PPG, by contrast, continues to rise with baseline expertise. The two measures therefore move together over most of the distribution but diverge sharply at the top. Low-expertise occupations have modest values of both; occupations in the middle and upper-middle of the distribution combine sizable productivity gains with large expansions in their qualified pool; and the highest-expertise occupations have the largest potential productivity gains but little expansion in their qualified pool.

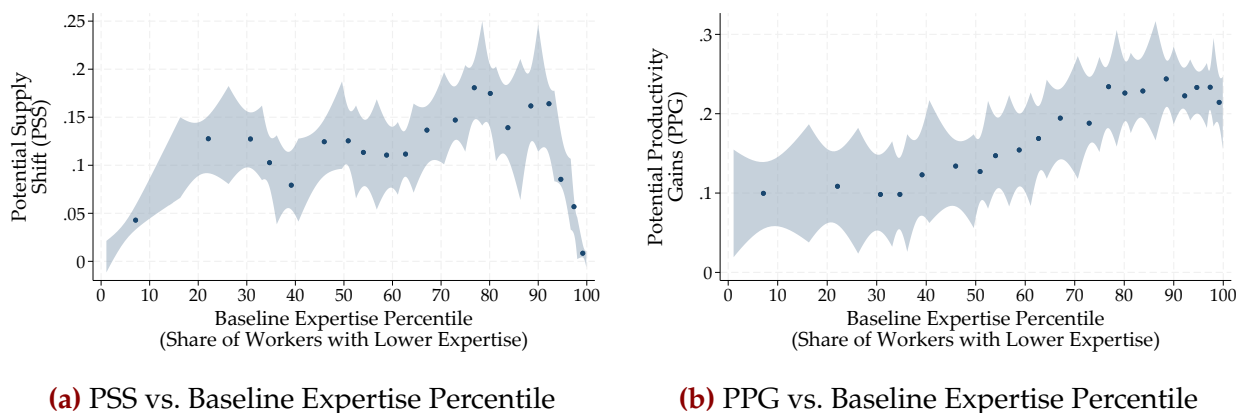


Figure 3: PSS and PPG vs. Baseline Expertise—Bin-scatter Plots

Notes: Bin-scatter plots at the occupation level. Occupations are sorted by their baseline expertise percentile, $100 \times F(X_o^{noAI})$, and grouped into 20 equal-sized bins; each point plots the bin mean of the y-axis variable against the bin mean of the percentile. Panel (a) plots PSS and Panel (b) plots PPG. Shaded areas are 95 percent confidence bands.

4.4 Benchmarking PSS Against Exposure Measures

A standard measure of GenAI’s reach is occupational “exposure,” which summarizes how strongly GenAI applies to an occupation’s tasks (e.g., [Eloundou et al., 2024](#); [Felten et al., 2023](#); [Gmyrek et al., 2025](#)) and underlies much of the research and policy discussion (e.g., [Goldman Sachs, 2023](#); [OECD, 2024](#); [IMF, 2025](#); [Brynjolfsson et al., 2025a](#); [The Budget Lab, 2025](#); [Hosseini and Lichtinger, 2025](#)). While exposure is commonly used to study the labor-demand effects of GenAI, it is less suited to capturing its supply-side effects for two reasons. First, it does not distinguish between automating easier and harder tasks, which can move the requirement of the remaining task bundle in opposite directions. Second, it misses the intensive margin: the reduction in expertise required for tasks that remain human-performed. PSS captures both margins and, importantly, translates the resulting change in an occupation’s requirement into the share of workers who become qualified.

Figure 4 plots PSS against the share of automatable tasks (Section 4.1.2). The two are positively related at low and moderate levels of exposure, but the relationship flattens among highly exposed occupations. More importantly, the dispersion is large throughout: exposure explains only 6 percent of the variation in PSS across occupations. Exposure alone is therefore an incomplete guide to how GenAI changes entry barriers and potential labor supply.

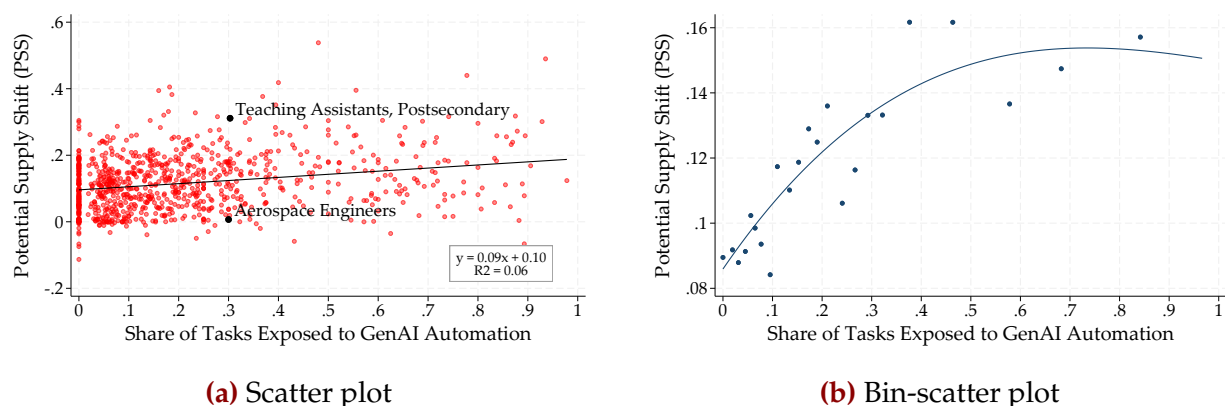


Figure 4: Automation Exposure vs. PSS

Notes: Each dot in Panel (a) is an occupation. Panel (b) groups occupations into equal-sized bins by their share of automatable tasks and plots mean PSS in each bin.

To make this dispersion concrete, Appendix Figure B.7 reports PSS for 32 occupations

with almost identical automatable shares—between 29 and 31 percent, around the 75th percentile of the exposure distribution. Their PSS ranges from 0.01 to 0.31. At the two ends are Aerospace Engineers and Postsecondary Teaching Assistants, both with 30 percent of tasks automatable. In both occupations, the automated tasks are among the easier ones, so removing them alone raises the requirement: from 40 to 49 months for Aerospace Engineers and from 6.8 to 8.6 months for Teaching Assistants. In both cases, the intensive margin more than offsets this increase, lowering the requirement to 35 and 2.1 months, respectively. What differs is where these changes occur in the expertise distribution. The drop from 6.8 to 2.1 months spans a dense part of the distribution, so the qualified pool for Teaching Assistants grows by 31 percent of the workforce. The drop from 40 to 35 months, by contrast, occurs in the thin upper tail and barely changes the qualified pool for Aerospace Engineers.

5 Inequality Implications

This section combines the model of Section 3 with the measures of Section 4 to quantify how GenAI affects wage inequality across workers. Section 5.1 describes the calibration. Section 5.2 reports the model’s results for the supply channel alone, the productivity channel alone, and the two together. Section 5.3 shows how the outcome of the race between the two channels depends on the elasticity of substitution across occupations. Throughout, we compare long-run equilibria: the counterfactuals describe how wages, employment, and inequality can change once workers have adopted GenAI, learned to use it effectively, and reallocated across occupations (Section 6 illustrates how the timing depends on the speed of adoption). These effects are therefore unlikely to be clearly visible in the data yet.

5.1 Calibration and Implementation

We take baseline wages and employment for 2024 from the BLS Occupational Employment and Wage Statistics, matched to O*NET occupations.²² The expertise distribution F

²²O*NET splits 64 six-digit SOC codes into several detailed occupations, for which BLS reports no separate employment, so we split each SOC employment total equally across them. Collapsing these occupa-

is the employment-weighted proxy from Section 4.1, which assigns all workers in an occupation that occupation’s requirement and therefore abstracts from within-occupation differences in expertise.²³ Appendix C.4 reports the counterfactuals under an alternative distribution constructed from CPS educational attainment.

Two parameters determine how the model responds to shocks: the elasticity of substitution across occupational outputs, σ , and the dispersion of occupational preferences, τ . The elasticity σ is central because it governs the relative strength of the productivity and supply channels (Section 3.5). There is no direct estimate of this elasticity for detailed occupations. Estimates for broad skill groups and broad occupation groups range from about 1.3 to 2 (Katz and Murphy, 1992; Acemoglu and Autor, 2011; Burstein et al., 2019; Caunedo et al., 2023), and detailed occupations are plausibly closer substitutes than broad groups. In the baseline, we set $\sigma = 4$, the value used in the numerical illustrations of Autor and Thompson (2025). However, because the results depend strongly on σ , we report results for values between 1.5 and 8 in Section 5.3 and show that they vary substantially across this range. We calibrate τ so that the model’s baseline employment shares, evaluated at observed wages and baseline requirements, are as close as possible to observed shares, which gives $\tau = 0.83$. Given σ and τ , we choose the occupation-specific shifters $\{B_o\}$ so that observed baseline wages are an exact fixed point given the model’s baseline employment and feasibility.²⁴ Table 2 summarizes the parameters.

For each counterfactual, we solve for the general equilibrium: workers reallocate across feasible occupations, and wages adjust until every occupational labor market clears. We compute inequality statistics across workers, weighting each occupation by its model employment in the equilibrium considered. The baseline variance of log wages is 0.164 under model employment and 0.229 under observed employment.

tions to the SOC level instead gives a similar supply effect and somewhat smaller productivity effects.

²³We drop two occupations from the counterfactuals, which therefore cover 883 occupations. For Urologists, the post-GenAI requirement (77 months) is above the highest baseline requirement (73 months), so under this proxy no worker qualifies after the shock and the model wage rises mechanically. Orthopedic Surgeons have the highest baseline requirement, so the proxy gives them no eligible workers even before the shock. The proxy F still includes the workers of both occupations. The mean PSS in this sample is 0.117, against 0.116 across the 889 occupations of Section 4.1.

²⁴Model employment then matches observed employment only loosely (the correlation across occupations is 0.18), because employment differs across occupations in the model only through wages and feasibility. The dynamic model in Section 6 matches observed employment exactly.

Table 2: Calibrated Parameters

Parameter	Description	Calibration	Value
σ	Elasticity of substitution across occupations	Set externally; varied from 1.5 to 8	4
τ	Dispersion of occupational preferences	Minimizes the employment-weighted distance between model and observed log employment shares	0.83
$\{B_o\}_{o \in \mathcal{O}}$	Occupation-specific shifters	Observed baseline wages are an exact fixed point, $w_o = B_o L_o^{-1/\sigma}$	One per occupation

5.2 Model Results

We report three counterfactuals: changes in expertise requirements alone (the supply channel), higher productivity alone (the productivity channel), and both together. For each single channel, we show how occupational wages change across the pre-AI wage distribution; Table 3 reports inequality across workers for all three.

5.2.1 The Supply Channel

We begin with the supply channel alone, the main focus of this paper. The counterfactual changes each occupation’s expertise requirement from R_o to $\tilde{R}_o$, as measured in Section 4.1, holds productivity fixed, and compares the resulting equilibrium with the baseline. Figure 5 plots the change in equilibrium occupational wages by each occupation’s percentile in the pre-AI wage distribution.

Wages fall in high-wage occupations and rise in low-wage ones. In the bottom quartile, wages rise by 5 to 10 percent as workers leave, while in the middle of the distribution they change by less than 5 percent. Above the 80th percentile, wages fall by 8 to 15 percent as workers who could previously enter only lower-barrier occupations gain access to these better-paid ones.²⁵

Table 3 turns from occupational wages to inequality across workers, which also depends on where workers are employed. Under the supply channel, the variance of log wages falls by 8.3 percent, from 0.164 to 0.150, and inequality in the lower half falls: the

²⁵These are changes between long-run equilibria. Section 6 examines the transition path and how long these effects may take to materialize.

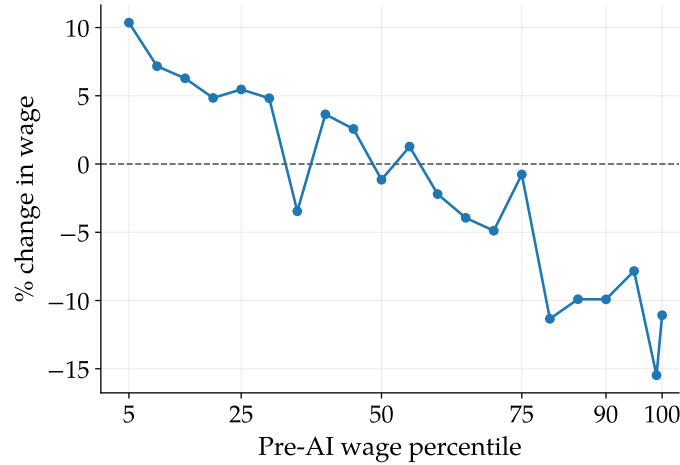


Figure 5: Wage Effects of the Supply Channel by Pre-AI Wage Percentile

Notes: Percentage change in equilibrium occupational wages when expertise requirements change, with productivity held fixed. Occupations are grouped into bins of the pre-AI wage distribution defined on observed employment, and changes are averaged within each bin with observed employment weights; the top bin covers the top one percent of employment.

$p_{50} - p_{10}$ gap narrows from 0.348 to 0.304. The $p_{90} - p_{50}$ gap, by contrast, remains nearly unchanged (0.658 to 0.662): wages fall in most high-wage occupations, but many more workers are now employed in them. Holding the allocation of workers fixed, the compression of occupational wage premia alone would reduce the variance by 21 percent; workers moving into better-paid occupations offset most of this effect (Appendix C.1).

PSS as a summary statistic: We next ask how well our empirical PSS measure captures the model’s occupational wage response. This matters because, like standard exposure measures, PSS can be computed directly from task-level data (Section 4.1), without solving the model. If it also predicts the wage changes generated by the model, it can serve as a simple occupation-level summary of GenAI’s supply-side effect. Figure 6 plots each occupation’s equilibrium wage change against its PSS. Occupations with larger PSS experience larger wage declines, and the relationship is close to linear (slope -0.96): PSS alone explains 48 percent of the employment-weighted variation in wage changes.

The fit is imperfect for two reasons. First, an occupation’s wage can change even when its PSS is close to zero. This is clearest for the lowest-requirement occupations, such as dishwashers: almost every worker already qualifies for them, so their PSS is near zero,

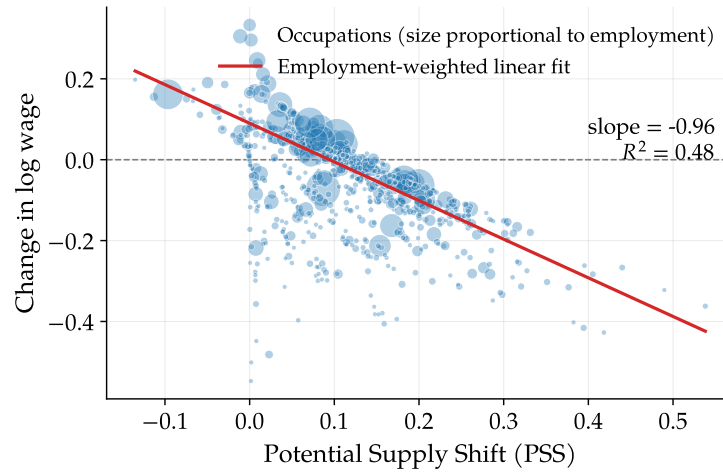


Figure 6: General-Equilibrium Wage Changes and the Potential Supply Shift

Notes: Each dot is an occupation, with area proportional to its employment. The vertical axis shows the general-equilibrium change in the occupation's log wage under the supply channel; the line is the employment-weighted linear fit.

but their wages rise because many of their workers move to occupations that GenAI has made accessible. Second, PSS measures newly eligible workers as a share of the whole workforce, while an occupation's wage responds to the size of this inflow relative to its baseline employment (Equation (16)). A given PSS is therefore a much larger supply shock for occupations with few eligible workers, typically those with high requirements, so their wages fall more than PSS alone predicts.

5.2.2 The Productivity Channel

We next consider productivity gains alone: each occupation's productivity rises from A_o to $A_o e^{\pi_o}$, with $\pi_o = -\log(1 - PPG_o)$ (Section 4.2), while expertise requirements remain at their baseline values. Figure 7 shows the resulting changes in occupational wages. Wages rise in all occupations, and above the 80th percentile they rise by about twice as much as in the bottom quartile, because PPG is larger in high-expertise, high-wage occupations (Figure 3b). Inequality across workers therefore rises: the variance of log wages increases by 25 percent, from 0.164 to 0.204, and both the lower- and upper-tail gaps widen (Table 3). Because expertise requirements do not change, there is relatively little reallocation, so al-

most all of the increase reflects wider wage gaps across occupations rather than changes in where workers are employed.

5.2.3 The Overall Effect on Inequality: The Race

When both channels operate, inequality across workers rises, but by less than under the productivity channel alone. The variance of log wages increases by 13 percent, from 0.164 to 0.185, compared with 25 percent under productivity gains alone (Table 3). The supply channel mainly dampens the increase in lower-tail inequality: the $p50 - p10$ gap widens by 8 percent, compared with 25 percent under the productivity channel alone. The $p90 - p50$ gap, by contrast, widens more (13 percent versus 11 percent), because more workers move into high-wage occupations. The rise in the variance of log wages comes primarily from this reallocation: holding the allocation of workers fixed, wage gaps between occupations narrow slightly.²⁶ Overall, at $\sigma = 4$ productivity therefore wins the race across workers and inequality increases. However, as shown by Section 5.3, the outcome depends strongly on σ and can reverse at lower, but still plausible, elasticities.

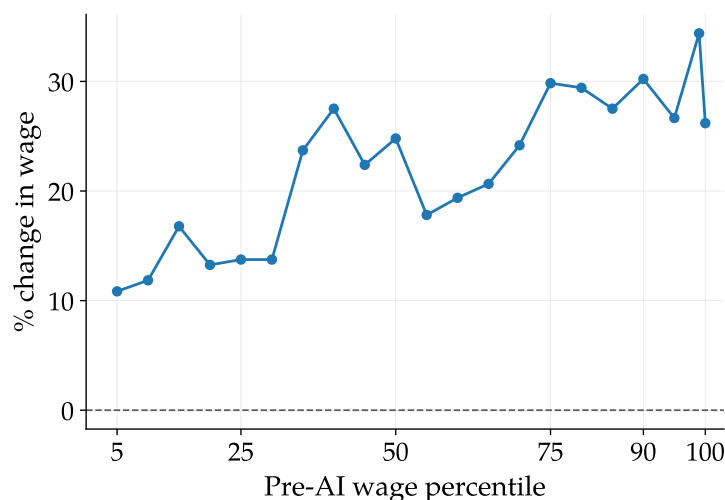


Figure 7: Wage Effects of the Productivity Channel by Pre-AI Wage Percentile

Notes: Percentage change in equilibrium occupational wages when occupational productivity rises with PPG, with expertise requirements held at baseline. Bins as in Figure 5.

²⁶Holding the allocation of workers across occupations fixed, the new occupational wages would lower the variance of log wages by 4 percent of its baseline value. The variance rises overall because workers move into higher-paid occupations, which adds 17 percent (Appendix C.1).

Table 3: Wage Inequality Under the GenAI Counterfactuals

Scenario	$\text{Var}(\log w)$	$p90 - p10$	$p50 - p10$	$p90 - p50$
Baseline	0.164 (0.0)	1.006 (0.0)	0.348 (0.0)	0.658 (0.0)
Supply channel	0.150 (−8.3)	0.965 (−4.0)	0.304 (−12.7)	0.662 (+0.5)
Productivity channel	0.204 (+24.9)	1.166 (+15.9)	0.436 (+25.2)	0.731 (+11.1)
Both channels	0.185 (+13.1)	1.115 (+10.9)	0.375 (+7.8)	0.740 (+12.5)

Notes: Variance and percentile gaps of log occupational wages across workers, in general equilibrium at $\sigma = 4$. Each scenario weights occupations by the model’s employment in that scenario. Numbers in parentheses are percent changes relative to the pre-AI baseline. The “Supply channel” changes expertise requirements, the “Productivity channel” raises occupational productivity, and “Both channels” combine the two.

5.3 The Role of σ in the Race

The overall effect on inequality depends strongly on the elasticity of substitution across occupations, σ . Our baseline sets $\sigma = 4$, but, as discussed in Section 5.1, there is no direct estimate at this level of occupational detail and a range of values is plausible. We therefore report results across a wide range of elasticities.

The elasticity matters because it determines how strongly wages respond to each channel (Section 3.5). When σ is high, buyers readily switch between occupational outputs, so a shock to an occupation’s production (either to its productivity or to its number of workers) mainly changes how much it sells. A productivity gain then raises the value of each worker’s output, and hence the wage, by more (Equation (5)). Since these gains are largest in high-wage occupations, the productivity channel widens inequality by more. Likewise, σ determines how steeply wages decline as an occupation employs more workers: the higher σ , the less concave output is in the number of workers, and the more slowly wages decline (Equation (4)). When workers move from low-wage to high-wage occupations, wages therefore fall by less in the occupations they enter and rise by less in those they leave, so the supply channel compresses inequality by less.

Consistent with this intuition, Table 4 shows that when σ is high, the supply channel reduces inequality across workers by less—and even raises it slightly at extremely high

elasticities ($\sigma = 8$)—while the productivity channel increases inequality by more. When σ is low, the supply channel reduces inequality by more and the productivity channel raises it by less. As a result, with both channels, inequality across workers rises by 26 percent at $\sigma = 6$ but falls by 9 percent at $\sigma = 2$. The threshold is about $\sigma = 2.7$: inequality rises above it and falls below it. Thus, over most of the plausible range, the two channels push inequality in opposite directions, and which one dominates depends on σ .²⁷

Table 4: Change in Inequality Across Workers by Elasticity of Substitution

σ	1.5	2	3	4	5	6	8
Supply channel	−21.5	−18.2	−12.6	−8.3	−5.1	−2.7	+0.9
Productivity channel	+7.8	+13.2	+20.3	+24.9	+28.0	+30.4	+33.6
Both channels	−17.1	−9.4	+3.5	+13.1	+20.2	+25.7	+33.5

Notes: Percent change in the employment-weighted variance of log occupational wages across workers between the pre-AI baseline and each general-equilibrium counterfactual. $\{B_o\}$ is recalibrated for each σ so that baseline wages remain an exact fixed point. The baseline is $\sigma = 4$ (Table 3).

6 Dynamic Extension

This section extends the static model in two directions. The first concerns the structure of entry barriers. The static model treats them as a single expertise threshold that any sufficiently skilled worker can cross, regardless of their current occupation. In practice, switching costs depend on the pair of occupations—moving from nursing to medicine is far easier than moving from nursing to law. We therefore let forward-looking workers face occupation-to-occupation retraining costs and ask whether lower entry barriers still compress wage inequality when the ease of entering an occupation depends on where a worker starts. The second extension concerns timing. Because the static model compares steady states only, we also use the dynamic model to illustrate how quickly the effects

²⁷ Appendix C.2 examines sensitivity to the size of productivity gains, since the time savings underlying PPG (50 and 25 percent on exposed tasks) are a calibration choice: realized savings could be larger, or smaller once adoption and checking costs are taken into account. Larger productivity gains lower the threshold, while smaller gains raise it. Appendix C.3 reports results for alternative values of ρ . The closed-form race condition in Appendix A.2, which concerns the expertise premium rather than inequality across workers, switches sign at about $\sigma = 5.8$. Under the CPS-based expertise distribution, the threshold for inequality across workers is about $\sigma = 2.9$ (Appendix Table C.7).

might unfold. Since the supply channel is the main focus of the paper, we isolate it here and abstract from productivity gains.

6.1 Model

Time is discrete and measured in years. There are J occupations. Workers are infinitely lived, discount the future at rate $\beta \in (0, 1)$, and are characterized only by their current occupation s ; as in the static model, all workers in an occupation earn the same wage, w_o . Each year, a worker in occupation s chooses an occupation o , possibly s itself, and the value of being in s satisfies

$$V(s) = \tau \log \sum_{o=1}^J \exp \left[\frac{\log w_o - c_{so} \mathbf{1}_{o \neq s} + a_o + \beta V(o)}{\tau} \right], \quad (22)$$

where a_o is a time-invariant amenity, c_{so} is the cost of switching from s to o , and $\tau > 0$ is the dispersion of i.i.d. Type-I extreme value taste shocks. The parameter τ plays the same role as in the static model but is calibrated separately (Section 6.3).

We let the switching cost scale with the retraining distance d_{so} , the time a worker in s needs to retrain for o , and allow an origin and a destination component:

$$c_{so} = (\kappa_s^{\text{out}} + \kappa_o^{\text{in}}) d_{so}. \quad (23)$$

The origin parameter $\kappa_s^{\text{out}} \geq 0$ captures how costly it is to leave s (for example, because of occupation-specific human capital), and the destination parameter $\kappa_o^{\text{in}} \geq 0$ captures how costly it is to enter o (for example, because of licensing). Together they let the model match how many workers leave each occupation and which occupations they enter, rather than only the aggregate mobility rate. The probability of moving from s to o is

$$P(o | s) = \frac{\exp[v(o | s)/\tau]}{\sum_{o'=1}^J \exp[v(o' | s)/\tau]}, \quad v(o | s) \equiv \log w_o - c_{so} \mathbf{1}_{o \neq s} + a_o + \beta V(o). \quad (24)$$

In a steady state, employment is the stationary distribution of this Markov chain, $\mu(o) = \sum_s P(o | s) \mu(s)$, labor supply is $L_o = \mu(o)$, and wages satisfy the same CES inverse de-

mand as in the static model, $w_o = B_o L_o^{-1/\sigma}$. Along a transition path, the same conditions hold year by year, with employment evolving as $\mu_{t+1}(o) = \sum_s P_t(o | s) \mu_t(s)$.

AI shock: We model the supply-side effect of GenAI as a permanent reduction in retraining distances, $d_{so} \rightarrow d_{so}^{\text{AI}}$: access to GenAI makes it easier for a worker in s to acquire the skills needed in o , more so for some pairs than for others. The switching-cost parameters ($\kappa^{\text{out}}, \kappa^{\text{in}}$), the dispersion τ , and the amenities $\{a_o\}$ are held fixed, and so are the productivity shifters $\{B_o\}$, except for the common term Γ , which rises with aggregate output and scales all wages equally (Section 3.4). Lower retraining distances are therefore the only channel. This is the dynamic counterpart of the static supply channel: there, GenAI shifts a single expertise threshold per occupation; here, it lowers bilateral retraining frictions. The reduction need not happen at once: it materializes as workers adopt GenAI and learn to use it effectively, which takes time. We capture this with an adoption period of T years, over which the reduction phases in linearly, $d_{so,y} = d_{so} + \phi_y (d_{so}^{\text{AI}} - d_{so})$ with $\phi_y = \min(y/T, 1)$ in year $y = 1, 2, \dots$, so that $T = 0$ is an immediate reduction. Because the speed of adoption is uncertain, we report results for T between 0 and 25 years. The shock is unanticipated; once it starts, workers know the future path of retraining distances.

6.2 Data Inputs

The first input is a directed retraining-distance matrix d_{so} : for each ordered pair of occupations, the time in years a worker in s needs to retrain for o . An LLM (GPT-5 Nano) estimates this time for each pair of detailed occupations based on their O*NET task bundles (Appendix B.1.5 reports the prompts), and we average these estimates within each ordered pair of 3-digit SOC groups ($J = 94$). The matrix is asymmetric—moving from nursing assistant to physician takes far longer than the reverse, because the physician already has most of the skills the nursing assistant needs—and has zeros on the diagonal.

To validate the matrix, we construct an independent benchmark from O*NET proficiency levels on 35 skill and 33 knowledge dimensions. The directed skill shortfall is $d_{so}^{\text{O*NET}} = \sum_{k=1}^{68} \max(0, v_o^k - v_s^k)$, where v_o^k is the proficiency level occupation o requires on dimension k . This measure captures the additional proficiency a worker in s needs to

meet the requirements of o . Across the 8,742 off-diagonal pairs, the two measures have a Pearson correlation of 0.49 and a Spearman correlation of 0.65 (Appendix Figure D.1), even though one is a holistic estimate of retraining time and the other a mechanical sum of skill gaps.

The second input is the post-AI matrix d_{so}^{AI} , constructed using the same procedure under the counterfactual that the worker has access to a capable GenAI assistant. The average off-diagonal retraining distance falls from 1.15 to 0.89 years, a decline of 22 percent (from 1.17 to 0.91 years when pairs are weighted by origin and destination employment).

The third input is observed occupational mobility from the Occupational Transitions Public Data Set of Schubert et al. (2024), which reports origin–destination shares of occupational switchers at the 6-digit SOC level using a large longitudinal sample of U.S. workers. We aggregate to the 3-digit level by summing switcher counts and treating moves within a 3-digit group as stays. This gives two sets of moments: the annual outflow rate from each origin, m_s^{data} , rescaled so that its employment-weighted mean equals the 12 percent annual occupational switching rate in the CPS, and the employment-weighted destination share of switchers, $f_o^{\text{data}} = \sum_s \omega_s \hat{P}(o \mid s, \text{switch})$, where $\omega_s = L_s^{\text{data}} / \sum_{s'} L_{s'}^{\text{data}}$ is the employment share of origin s .

6.3 Calibration

We set $\sigma = 4$, as in the static model, and $\beta = 0.95$. Amenities $\{a_o\}$ are recovered by a contraction mapping so that pre-AI employment matches the data, and productivity shifters by $B_o = w_o^{\text{data}} L_o^{1/\sigma}$, so the pre-AI steady state reproduces observed wages and employment exactly. We set the dispersion of taste shocks to $\tau = 0.46$. Because the model reproduces observed wages and employment by construction, our moments do not separately identify τ : observed mobility pins down switching costs relative to τ , while τ itself governs how strongly workers respond to the wage changes induced by the shock. Appendix D.2 reports results for $\tau = 0.25$ and for $\tau = 0.83$, the value calibrated in the static model.²⁸ Holding τ fixed, the $2J$ switching-cost parameters $(\kappa^{\text{out}}, \kappa^{\text{in}})$ are chosen by a fixed-point iteration to match the J outflow rates and the J destination shares. The model

²⁸The value $\tau = 0.46$ comes from an earlier version of the calibration, with a common switching cost for all pairs of occupations.

matches these moments: the correlation between model and data is 0.98 for outflow rates and 0.999 for destination shares.²⁹ Appendix Table D.1 summarizes the details.

6.4 Results

Figure 8 shows the transition paths for adoption periods between 0 and 25 years.³⁰ The long-run outcome is the same for every adoption path: wage inequality across workers falls, with the variance of log wages declining from 0.208 to 0.188, or by 9.8 percent. The size of this decline depends on τ , which our moments do not pin down: it is 5 percent at $\tau = 0.25$ and 17 percent at $\tau = 0.83$ (Appendix D.2). The direction matches the static supply channel, and at our baseline τ the magnitude is similar to the static decline of 8.3 percent (Table 3), although the two estimates are not directly comparable.³¹

The speed of the transition depends mainly on how quickly GenAI is adopted. After a decade, 89 percent of the long-run decline in inequality has been realized if retraining distances fall immediately, 63 percent if adoption takes 10 years, and only 18 percent if it takes 25 years. Thus, while only illustrative, these results suggest that GenAI’s supply channel can lower wage inequality, but that its effects may unfold over decades rather than years if adoption is gradual.³²

Finally, we examine the reallocation across occupations underlying the long-run decline in inequality (Figure 9). The occupations that expand most are high-wage cognitive occupations: Computer Occupations (+1.6 million workers, +33 percent), Other Management (+1.4 million, +41 percent), Lawyers and Judges (+0.9 million, +109 percent), and Financial Specialists (+0.8 million, +28 percent). Their wages fall by 4 to 16 percent as

²⁹The fixed-point iteration does not converge exactly: for a few occupations the model’s flows remain off target by a factor of about three to four, and some origin parameters κ_s^{out} reach their lower bound of essentially zero, so these targets cannot be matched with non-negative switching costs. The model’s aggregate annual switching rate is 10.5 percent, against the 12 percent target.

³⁰We solve each transition path by backward induction on the value function given a guessed wage path, forward propagation of the employment distribution from the pre-AI steady state, and damped iteration on wages until labor markets clear in every year.

³¹The dynamic model uses 94 broad occupation groups rather than 883 detailed occupations, makes entry costs depend on the worker’s origin occupation, and matches observed employment exactly.

³²These shares change little with τ : for τ between 0.25 and 0.83, they range from 82 to 94, 58 to 69, and 17 to 19 percent, respectively (Appendix D.2). Two simplifications also matter for timing. Because workers are infinitely lived, the model likely overstates the speed of adjustment: older workers would have less incentive to retrain. Because it has no new cohorts, the model likely understates it: new entrants could move directly into newly accessible occupations without paying switching costs.

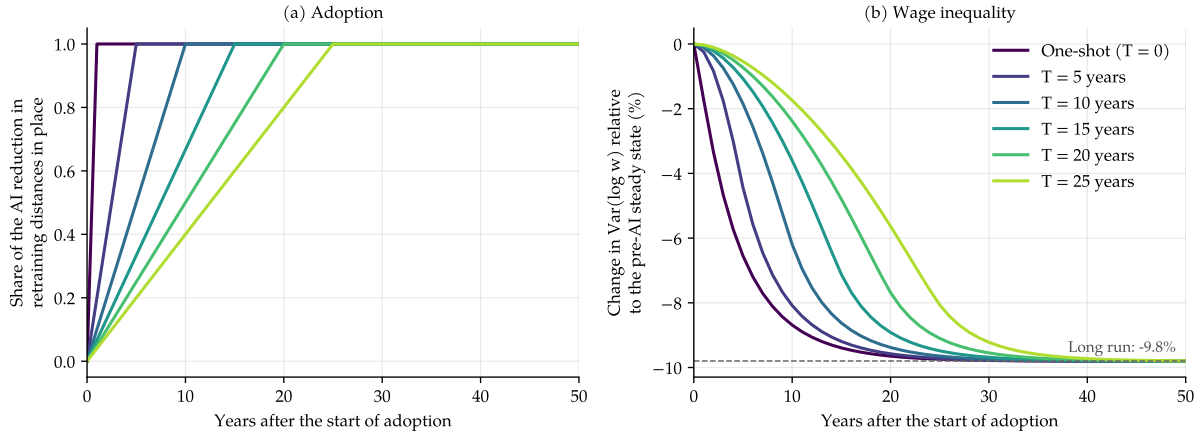


Figure 8: Wage Inequality Over Time, by Adoption Speed (Supply Channel)

Notes: Panel (a) shows the share of the AI-induced reduction in retraining distances in place, $\phi_y = \min(y/T, 1)$, for adoption periods T from 0 (immediate adoption) to 25 years. Panel (b) shows the percentage change in the variance of log wages relative to the pre-AI steady state, weighting occupations by employment in each year; the dashed line marks the long-run change (−9.8 percent). Switching-cost parameters, amenities, and productivity shifters are held fixed.

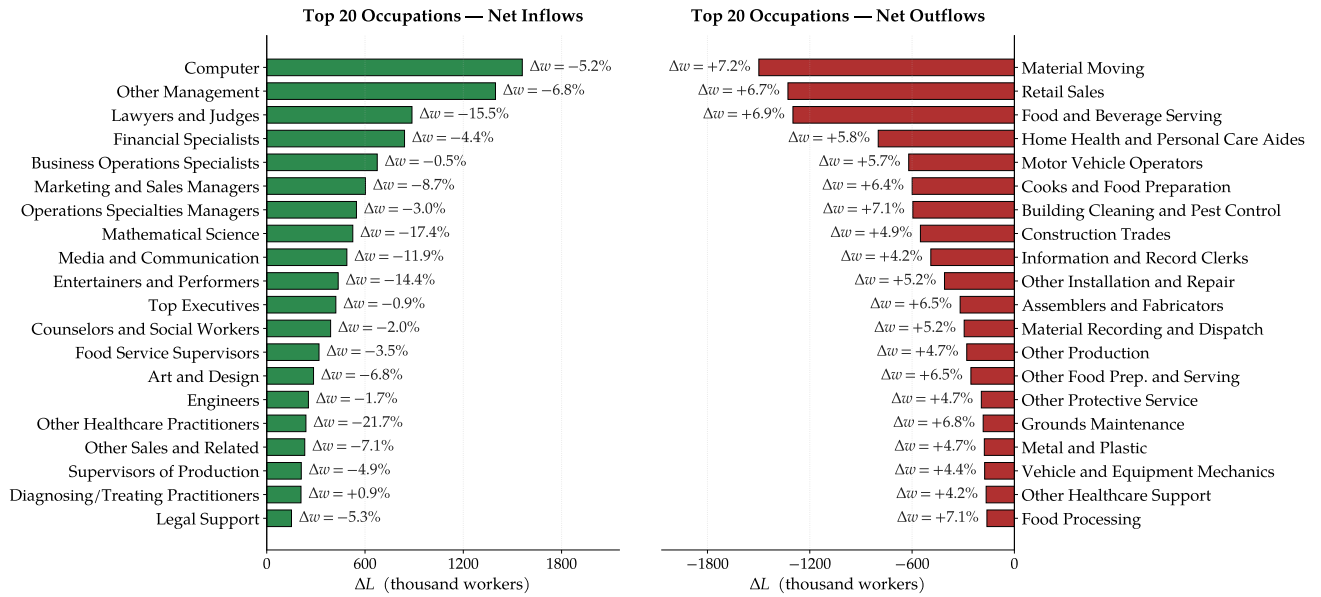


Figure 9: Occupation-Level Employment Reallocation (Supply Channel)

Notes: The left panel shows the twenty occupations with the largest employment inflows between the pre- and post-AI steady states; the right panel shows the twenty with the largest outflows. Bars report employment changes in thousands of workers (ΔL), and labels report percentage changes in occupational wages (Δw).

scarcity premia erode. The occupations that shrink most are low-wage service and manual occupations: Material Moving (−1.5 million, −19 percent), Retail Sales (−1.3 million, −17 percent), Food and Beverage Serving (−1.3 million, −18 percent), and Home Health and Personal Care Aides (−0.8 million, −15 percent). Their wages rise by 6 to 7 percent as outflows reduce labor supply. These patterns suggest that GenAI’s supply-side effects could substantially reshape the occupational structure, although Figure 8 illustrates that these changes may unfold gradually over many years.

7 Conclusion

Discussions of how GenAI will affect occupations often focus on labor demand: which occupations it is likely to augment or automate the most. This paper studies how GenAI can affect wage inequality through a labor-supply channel: by changing the expertise needed to perform an occupation’s tasks, GenAI can shift the pool of workers who can enter the occupation and, with it, the occupation’s scarcity premium. To measure this channel, we construct the Potential Supply Shift (PSS) index, which we make publicly available. PSS translates LLM-based assessments of how GenAI changes task-level expertise requirements—through tasks it automates and tasks that remain human-performed but become easier—into the change in the share of workers qualified for each occupation. In our measures, requirements fall on average, mostly because the remaining tasks become easier rather than because of automation. PSS rises with an occupation’s expertise requirement but falls at the very top, where few workers are close to the threshold, and it is only weakly related to standard exposure measures.

Embedding PSS and a companion measure of Potential Productivity Gains (PPG) in a general-equilibrium model shows that the two channels pull in opposite directions. Lower entry barriers compress occupational wage premia, drawing workers from low-wage into high-wage occupations, while productivity gains, which are largest in high-wage occupations, widen them. Which force dominates inequality across workers depends on the elasticity of substitution across occupations: a higher elasticity weakens the supply channel and strengthens the productivity channel, because an inflow of workers then depresses an occupation’s wage less and a productivity gain raises it more. With both channels, inequality across workers falls by 9 percent at an elasticity of 2 but rises

by 26 percent at 6, with a threshold of about 2.7 (at this level of occupational detail, the elasticity has not been estimated). PSS alone accounts for about half of the variation in the model's occupational wage changes under the supply channel, so it offers a simple guide, computable from task data, to where GenAI's supply effects are likely to be largest.

A dynamic extension relaxes the single expertise threshold: forward-looking workers face retraining costs that depend on both their current occupation and their destination, and GenAI lowers the underlying retraining distances. Lower entry barriers still reduce wage inequality across workers, as employment shifts toward high-wage cognitive occupations and their wages fall. The extension also highlights timing: because adoption and reallocation take time, these changes may unfold over decades rather than years if adoption is gradual.

Several caveats apply. The LLM-based assessments of how GenAI changes task expertise are difficult to validate directly, and we hold the set of tasks and occupations fixed, abstracting from new tasks and the rebundling of existing ones. Because the supply channel requires adoption and worker reallocation, it is also unlikely to be clearly visible in current data. With these caveats, our results suggest that GenAI's distributional effects cannot be read from productivity gains or exposure alone: by changing who can enter which occupations, GenAI can also reshape the scarcity premia that occupations earn. As adoption proceeds, a central empirical question is whether GenAI expands entry into high-expertise occupations and erodes their scarcity premia.

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Supplemental Appendix

A Model and Theory

A.1 Analytical Characterization of the Expertise Premium

This appendix characterizes how GenAI affects the *expertise premium*, the elasticity of wages with respect to occupational expertise requirements, in two special cases of the model that deliver closed-form results: an economy with two occupations, and a continuum of occupations with Pareto-distributed expertise and power-law demand. These assumptions are more restrictive than those of the model in Section 3, which we solve numerically in Section 5, but they show transparently which primitives determine whether GenAI steepens or flattens the premium. The expertise premium is related to, but distinct from, wage inequality across workers, which also depends on how workers are distributed across occupations; the end of this appendix discusses the difference.

A.1.1 Two-Occupation Model

Setup: There are two occupations, Low (L) and High (H), with expertise requirements $R_L = 0 < R_H$. Workers draw expertise e from a continuous distribution F with density f . Let $m_L := F(R_H)$ be the mass of workers qualified for L only and $m_H := 1 - F(R_H)$ the mass qualified for both occupations, so that $m_L + m_H = 1$. Labor demand is $w_o = B_o L_o^{-1/\sigma}$ for $o \in \{L, H\}$, with $\sigma > 1$. Workers with $e < R_H$ can work only in L ; workers with $e \geq R_H$ choose between L and H according to the logit model of Section 3.3, with dispersion $\tau > 0$. Defining $q := (w_H/w_L)^{1/\tau}$, labor supply is

$$L_H = m_H \frac{q}{1+q}, \tag{A.1}$$

$$L_L = m_L + m_H \frac{1}{1+q}. \tag{A.2}$$

Equilibrium: The equilibrium relative wage $x^* := w_H/w_L$, the two-occupation counterpart of the expertise premium, is the unique solution to

$$x = \frac{B_H}{B_L} \left(\frac{m_H q}{m_L(1+q) + m_H} \right)^{-1/\sigma}, \quad q = x^{1/\tau}. \quad (\text{A.3})$$

Taking the ratio of the two demand equations and substituting Equations (A.1)–(A.2) yields Equation (A.3). Denote its right-hand side by $\Psi(x)$. As $x \rightarrow 0$, $\Psi(x) \rightarrow \infty$, and as $x \rightarrow \infty$, $\Psi(x) \rightarrow (B_H/B_L)(m_H/m_L)^{-1/\sigma} < \infty$. Since Ψ is continuous and strictly decreasing, it crosses the identity exactly once. Under perfect sorting ($\tau \rightarrow 0$), and provided that $w_H > w_L$, all workers qualified for H work in H :

$$L_L = m_L, \quad L_H = m_H, \quad w_o = B_o m_o^{-1/\sigma}, \quad x^* = \frac{B_H}{B_L} \left(\frac{m_H}{m_L} \right)^{-1/\sigma}. \quad (\text{A.4})$$

GenAI shocks: The productivity channel shifts the ratio of the demand shifters. Since $B_o \propto A_o^{1-1/\sigma}$ and $A_o^{\text{AI}} = A_o \exp(\pi_o)$, with $\pi_o \approx PPG_o$ (Section 3.4),

$$\Delta \log \frac{B_H}{B_L} \approx \frac{\sigma-1}{\sigma} (PPG_H - PPG_L). \quad (\text{A.5})$$

The scarcity channel changes the mass of workers qualified for H :

$$PSS_H := F(R_H) - F(\tilde{R}_H) = dm_H. \quad (\text{A.6})$$

Proposition 1 (The productivity–scarcity race). *Under small simultaneous PPG and PSS shocks, the change in the relative wage is*

$$d \log x^* = \mathcal{D} \left[\underbrace{\frac{\sigma-1}{\sigma} (PPG_H - PPG_L)}_{\text{productivity}} - \underbrace{\frac{1}{\sigma} \frac{q}{1+q} \left(\frac{1}{L_H} + \frac{1}{L_L} \right) PSS_H}_{\text{scarcity}} \right], \quad (\text{A.7})$$

where

$$\mathcal{D} := \frac{1}{1 + \mathcal{E}/\sigma} \in (0, 1), \quad \mathcal{E} := \frac{\partial \log(L_H/L_L)}{\partial \log x} > 0.$$

$\mathcal{D}$ is a general-equilibrium dampening factor, and $\mathcal{E}$ is the elasticity of relative employment with respect to the relative wage, holding m_H fixed.

Proof. Totally differentiate $\log x = \log(B_H/B_L) - (1/\sigma) \log(L_H/L_L)$. Holding x fixed, Equations (A.1)–(A.2) imply $\partial \log(L_H/L_L)/\partial m_H = [q/(1+q)](1/L_H + 1/L_L)$. Adding the endogenous response of relative employment, $\mathcal{E} d \log x$, and solving for $d \log x$ yields Equation (A.7). $\square$

The relative wage rises when the PPG differential is large relative to the scarcity term and falls when PSS_H is large. Under perfect sorting, the two channels can also be traced to wage levels. Equation (A.4) gives $w_o^{\text{AI}} = B_o^{\text{AI}}(m_o^{\text{AI}})^{-1/\sigma}$, with $m_L^{\text{AI}} = m_L - PSS_H$ and $m_H^{\text{AI}} = m_H + PSS_H$, so that, omitting the change in the common term Γ , which scales both wages equally,

$$\Delta \log w_L = \underbrace{\frac{\sigma-1}{\sigma} PPG_L}_{\text{productivity}} + \underbrace{\frac{1}{\sigma} \log \frac{m_L}{m_L - PSS_H}}_{\text{scarcity: workers leave } L}, \quad (\text{A.8})$$

$$\Delta \log w_H = \underbrace{\frac{\sigma-1}{\sigma} PPG_H}_{\text{productivity}} - \underbrace{\frac{1}{\sigma} \log \frac{m_H + PSS_H}{m_H}}_{\text{scarcity: workers enter } H}. \quad (\text{A.9})$$

When $PSS_H > 0$, the scarcity channel raises w_L , because workers leave L , and lowers w_H , because new entrants erode its scarcity premium; both effects compress the relative wage. The productivity channel raises both wages, by amounts that depend on each occupation's PPG.

A.1.2 Continuum of Occupations with Pareto Expertise

Environment: Occupations form a continuum indexed by their baseline expertise requirement $R \in [\underline{R}, \infty)$, with $\underline{R} > 0$. Demand shifters follow a power law, $B(R) = b R^\gamma$ with $b > 0$ and $\gamma \geq 0$, and CES final demand implies $w(R) = B(R) \ell(R)^{-1/\sigma}$, where $\ell(R)$ is the employment density and $\sigma > 1$. Expertise is Pareto-distributed on $[\underline{R}, \infty)$:

$1 - F(e) = (\underline{R}/e)^\theta$ and $f(e) = \theta \underline{R}^\theta e^{-(\theta+1)}$, with $\theta > 1$.

Baseline equilibrium under perfect sorting: Under perfect sorting ($\tau = 0$), each worker chooses the highest-wage occupation she qualifies for. If wages increase with R (verified below), a worker with expertise e works in occupation $R = e$, so the employment density equals the expertise density:

$$\ell(R) = f(R) = \theta \underline{R}^\theta R^{-(\theta+1)}. \quad (\text{A.10})$$

Substituting into the wage equation gives $w(R) = b (\theta \underline{R}^\theta)^{-1/\sigma} R^{\gamma+(\theta+1)/\sigma}$, which increases with R and confirms the assignment. The expertise premium is therefore the same at every R :

$$\xi := \frac{d \log w}{d \log R} = \gamma + \frac{\theta + 1}{\sigma}. \quad (\text{A.11})$$

Wages rise with expertise both because demand rises with it (γ) and because qualified workers become scarcer $((\theta + 1)/\sigma)$.

GenAI shocks: The effect of GenAI on the premium depends on two gradients across occupations:

$$\mathcal{G}_{PPG} := \frac{d PPG(R)}{d \log R}, \quad \mathcal{G}_{PSS} := \frac{d \log(S^{\text{AI}}(R)/S(R))}{d \log R}, \quad (\text{A.12})$$

where $S(R) = 1 - F(R)$ and $S^{\text{AI}}(R) = 1 - F(\tilde{R}(R))$ are the baseline and post-GenAI eligible pools (Equation (8)) and $\tilde{R}(R)$ is the post-GenAI requirement of occupation R . $\mathcal{G}_{PPG}$ measures how fast productivity gains rise with expertise, and $\mathcal{G}_{PSS}$ how fast the *proportional* expansion of the eligible pool rises with expertise.

We assume that both gradients are constant. On the demand side, productivity gains are log-linear in the requirement, $\pi(R) = \pi_0 + \mathcal{G}_{PPG} \log(R/\underline{R})$, where we use $\pi \approx PPG$ to read the slope as the PPG gradient. Since $B \propto A^{1-1/\sigma}$, the demand exponent becomes $\gamma + (\sigma - 1)\mathcal{G}_{PPG}/\sigma$. On the supply side, GenAI lowers requirements according to $\tilde{R}(R) = \underline{R}^{1-\delta} R^\delta$, with $0 < \delta \leq 1$, so that the lowest requirement is unchanged and

higher requirements fall proportionally more. Then

$$S(R) = \left(\frac{\underline{R}}{R}\right)^\theta, \quad S^{AI}(R) = \left(\frac{\underline{R}}{R}\right)^{\delta\theta}, \quad \frac{S^{AI}(R)}{S(R)} = \left(\frac{\underline{R}}{R}\right)^{(1-\delta)\theta}, \quad (\text{A.13})$$

so that

$$\mathcal{G}_{PSS} = (1 - \delta)\theta. \quad (\text{A.14})$$

$\mathcal{G}_{PSS}$ is positive when GenAI compresses requirements ($\delta < 1$), and it depends on two primitives: the degree of compression, $1 - \delta$, and the thinness of the upper tail of the expertise distribution, θ .

Proposition 2 (Post-GenAI expertise premium). *Under Pareto expertise, power-law demand, log-linear productivity gains, the compression of requirements above, and perfect sorting ($\tau = 0$), the post-GenAI expertise premium is*

$$\zeta^{AI} = \underbrace{\gamma + \frac{\sigma - 1}{\sigma} \mathcal{G}_{PPG}}_{\text{demand gradient}} + \underbrace{\frac{\theta + 1 - \mathcal{G}_{PSS}}{\sigma}}_{\text{scarcity gradient}}, \quad (\text{A.15})$$

and the change in the premium is

$$\zeta^{AI} - \zeta = \frac{1}{\sigma} [(\sigma - 1) \mathcal{G}_{PPG} - \mathcal{G}_{PSS}]. \quad (\text{A.16})$$

Proof. After the shock, a worker with expertise e qualifies for occupations up to $R^*(e) = \underline{R}^{(\delta-1)/\delta} e^{1/\delta}$ and, if wages increase with R (verified below), works in occupation $R^*(e)$. Changing variables with $e = \tilde{R}(R)$ gives the post-GenAI employment density

$$\begin{aligned} \ell^{AI}(R) &= f(\tilde{R}(R)) \tilde{R}'(R) = \frac{\theta \underline{R}^\theta}{(\underline{R}^{1-\delta} R^\delta)^{\theta+1}} \delta \underline{R}^{1-\delta} R^{\delta-1} \\ &= \theta \delta \underline{R}^{\delta\theta} R^{-(\delta\theta+1)}, \end{aligned}$$

which integrates to one over $[\underline{R}, \infty)$. The post-GenAI demand shifter and inverse demand

give

$$B^{\text{AI}}(R) = B(R) \exp\left(\frac{\sigma-1}{\sigma} \pi(R)\right) \propto R^{\gamma+(\sigma-1)\mathcal{G}_{\text{PPG}}/\sigma},$$

$$w^{\text{AI}}(R) = B^{\text{AI}}(R) \ell^{\text{AI}}(R)^{-1/\sigma} \propto R^{\gamma+(\sigma-1)\mathcal{G}_{\text{PPG}}/\sigma+(\delta\theta+1)/\sigma},$$

which increases with R . Since $\mathcal{G}_{\text{PSS}} = (1-\delta)\theta$, the last term equals $(\theta+1-\mathcal{G}_{\text{PSS}})/\sigma$, which yields Equations (A.15) and (A.16). $\square$

Given these functional forms, the result is exact; the only approximation is $\pi \approx \text{PPG}$, which expresses the productivity gradient in terms of PPG. Equation (A.16) yields a race condition: the expertise premium steepens if and only if

$$(\sigma-1)\mathcal{G}_{\text{PPG}} > \mathcal{G}_{\text{PSS}}. \quad (\text{A.17})$$

PPG affects only the demand gradient and PSS only the scarcity gradient, and the two enter with opposite signs. Productivity dominates when productivity gains rise steeply with expertise ($\mathcal{G}_{\text{PPG}}$ large) and when occupational outputs are close substitutes (σ large), so that a productivity gain raises labor demand by more. Scarcity dominates when barrier compression is strong and the upper tail of the expertise distribution is thin ($\mathcal{G}_{\text{PSS}} = (1-\delta)\theta$ large), so that even modest reductions in requirements expand the eligible pool proportionally more at high expertise levels. Because the condition involves only σ and the two gradients, which are defined over observables, it can be evaluated in the data (Appendix A.2).

Imperfect sorting: With logit choice ($\tau > 0$), the expertise premium is approximately the same as under perfect sorting for all but the lowest-expertise occupations. Conjecture $w(R) = C(\tau)R^{\tilde{\zeta}}$ and let $\zeta := \tilde{\zeta}/\tau + 1$. A worker with expertise e chooses among occupations $R' \leq e$ with probabilities proportional to $w(R')^{1/\tau}$, so the inclusive value is $\Phi(e) = C^{1/\tau}(e^{\tilde{\zeta}} - \underline{R}^{\tilde{\zeta}})/\zeta$. Away from the bottom of the distribution ($R/\underline{R}$ bounded away from one), $\Phi(e) \approx C^{1/\tau}e^{\tilde{\zeta}}/\zeta$ for all $e \geq R$, and aggregating over eligible workers gives

$$\ell(R) \propto R^{\tilde{\zeta}/\tau} \int_R^\infty e^{-(\theta+1)} e^{-\tilde{\zeta}} de \propto R^{\tilde{\zeta}/\tau} R^{-(\theta+\zeta)} = R^{-(\theta+1)}.$$

Inverse demand then gives $\zeta = \gamma + (\theta + 1)/\sigma$, as under perfect sorting: the wage level $C(\tau)$ depends on τ , but the premium does not. The approximation error is of order $(\underline{R}/R)^\zeta$, with $\zeta > 1$, so it vanishes quickly as R rises above $\underline{R}$. The same argument applied to the post-GenAI economy yields ζ^{AI} in Proposition 2. This result reflects the self-similarity of the Pareto distribution: with power-law wages and Pareto expertise, logit choice preserves the power-law shape of employment for any τ .

The level of PSS: Under Pareto expertise and the compression above,

$$PSS(R) = S^{\text{AI}}(R) - S(R) = \left(\frac{R}{\underline{R}}\right)^{\delta\theta} - \left(\frac{R}{\underline{R}}\right)^\theta. \quad (\text{A.18})$$

PSS is hump-shaped in R : it is zero at $R = \underline{R}$, rises through the middle of the distribution and falls in the upper tail as both eligible pools vanish, as in Figure 3a. The proportional expansion $S^{\text{AI}}(R)/S(R)$, by contrast, increases monotonically, and it is its gradient, $\mathcal{G}_{PSS}$, that governs the expertise premium.

Expertise premium and inequality across workers: The expertise premium is the slope of log wages with respect to log requirements. Inequality across workers also depends on how workers are distributed across occupations, and lower barriers change that distribution by moving workers into higher-paid occupations. A flatter premium therefore need not mean lower inequality across workers. In the quantitative model, this composition effect offsets most of the compression of the wage structure: at $\sigma = 4$, the supply channel lowers the variance of log wages by 21.1 percent of its baseline value when the allocation of workers is held fixed, and the reallocation of workers adds back 12.7 percent. At $\sigma = 8$, the composition effect exceeds the wage-structure effect, and the supply channel raises inequality across workers slightly (Appendix C.1).

A.2 Empirical Assessment of the Analytical Race Condition

We complement the quantitative counterfactuals in Section 5.2 with a direct empirical assessment of the race condition derived in Appendix A.1. That characterization implies that the expertise premium steepens if and only if $(\sigma - 1) \mathcal{G}_{PPG} > \mathcal{G}_{PSS}$ (Equation (A.17)),

where $\mathcal{G}_{PPG}$ and $\mathcal{G}_{PSS}$ denote, respectively, the gradients of PPG and of the log proportional expansion in the eligible pool with respect to log expertise (Equation (A.12)). Because both gradients are defined over observables, the condition can be evaluated directly in the data.

We estimate $\mathcal{G}_{PPG}$ as the slope from an employment-weighted regression of occupation-level PPG_o on $\log R_o$, where R_o is the baseline expertise requirement (months of training without GenAI). We estimate $\mathcal{G}_{PSS}$ as the slope from an employment-weighted regression of $\log(S_o^{AI}/S_o)$ on $\log R_o$, where $S_o = 1 - F(R_o)$ and $S_o^{AI} = 1 - F(\tilde{R}_o)$ are the baseline and post-GenAI eligible pools defined in Section 4.1.³³

Table A.1 reports the estimates. In the baseline specification (employment-weighted, $\sigma = 4$), $\hat{\mathcal{G}}_{PPG} = 0.036$ and $\hat{\mathcal{G}}_{PSS} = 0.175$, both precisely estimated. The race condition yields $(\sigma - 1)\hat{\mathcal{G}}_{PPG} = 0.109 < 0.175 = \hat{\mathcal{G}}_{PSS}$: the scarcity force exceeds the productivity force, implying that GenAI *flattens* the expertise premium. The implied change is $\Delta\zeta = -0.017$, a decline of about 5 percent relative to the baseline empirical expertise premium ($\hat{\zeta} \approx 0.31$, estimated by regressing log wages on log expertise). To gauge the magnitude: across the $p90$ – $p10$ range of baseline log expertise (3.3 log points), $\Delta\zeta = -0.017$ narrows the log wage gap by about 0.06, which is small relative to the baseline $p90$ – $p10$ log wage gap of 1.25 across workers (observed employment weights). The sign of the race condition depends on σ : it favors scarcity at $\sigma = 4$ and productivity for σ above about 5.8. Unweighted OLS gives a larger scarcity gradient and $\Delta\zeta = -0.029$.

The closed-form race condition points toward a slight flattening of the expertise premium, whereas the quantitative counterfactuals in Section 5.2.3 show inequality across workers rising on net. The two are not in conflict. The expertise premium is the slope of the wage–expertise relationship across occupations; it is related to, though not the same as, the wage-structure effect in the decomposition of Appendix C.1, which holds the allocation of workers fixed. At $\sigma = 4$ the two point the same way: the premium falls by 0.017 here, and in the full model the wage-structure effect of both channels is –4 percent of baseline variance. The increase in inequality across workers comes from the reallocation of workers toward higher-paid occupations, which neither object registers. In both

³³Because GenAI raises the effective expertise requirement for a small number of occupations (about 5 percent of occupations, representing 5 percent of employment), $S_o^{AI} < S_o$ in those cases. These observations are included in all regressions; excluding them moves the estimates further toward scarcity ($\hat{\mathcal{G}}_{PSS} = 0.19$, $\Delta\zeta = -0.019$).

Table A.1: Empirical Assessment of the Productivity–Scarcity Race Condition

Specification	$\hat{\mathcal{G}}_{PPG}$	$\hat{\mathcal{G}}_{PSS}$	$(\sigma-1)\hat{\mathcal{G}}_{PPG}$	$\Delta\xi$
Employment-weighted	0.036 (0.003)	0.175 (0.007)	0.109	−0.017
Unweighted	0.039 (0.003)	0.233 (0.009)	0.117	−0.029

Notes: $\hat{\mathcal{G}}_{PPG}$ is the slope from regressing PPG_o on $\log R_o$; $\hat{\mathcal{G}}_{PSS}$ is the slope from regressing $\log(S_o^{AI}/S_o)$ on $\log R_o$. Conventional standard errors in parentheses. $\Delta\xi = [(\sigma-1)\hat{\mathcal{G}}_{PPG} - \hat{\mathcal{G}}_{PSS}]/\sigma$. $\sigma = 4$. $N = 887$: the 889 occupations with employment data except Orthopedic Surgeons and Urologists, whose baseline or post-GenAI eligible pool is empty.

exercises, which force dominates depends on σ .

A.3 Automation as a Demand-Side Channel

This appendix extends the benchmark model by introducing a classical automation/displacement margin on the demand side. The extension nests the benchmark model when automation services are prohibitively expensive or when the technology offers no scope for substitution away from labor. The goal is conceptual: to show how a task-based measure of automation *feasibility* enters the occupational labor-demand schedule, and to clarify which additional objects must be specified to quantify its equilibrium incidence.

A.3.1 Task Structure and Automation Services

Each occupation o consists of tasks $t \in T_o = \{1, \dots, J_o\}$ with O*NET weights $\varphi_{ot} > 0$ and normalized weights

$$\omega_{ot} := \frac{\varphi_{ot}}{\sum_{s \in T_o} \varphi_{os}}, \quad \sum_{t \in T_o} \omega_{ot} = 1.$$

Let $\ell_{ot} \geq 0$ denote labor time allocated to task (o, t) and let $k_{ot} \geq 0$ denote *automation services* (e.g., AI services or AI-enabled capital services) used in that task. We interpret k_{ot} as an input measured in efficiency units and priced at p_k , an *effective* unit cost that subsumes compute/model costs as well as complementary integration, supervision, and organizational adjustment. Labor is augmented by GenAI through π_o as in (11).

We model each task as a CES combination of effective labor and automation services:

$$y_{ot} = \left((1 - I_{ot})^{\frac{1}{\eta}} (e^{\pi_o} \ell_{ot})^{\frac{\eta-1}{\eta}} + I_{ot}^{\frac{1}{\eta}} k_{ot}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \quad I_{ot} \in [0, 1], \eta > 1. \quad (\text{A.19})$$

The parameter I_{ot} captures the scope for automation in task (o, t) . When $I_{ot} = 0$, the task is strictly labor-only; when $I_{ot} > 0$, the producer can substitute between labor and automation services depending on relative prices.

Task outputs are aggregated into occupational output via a CES aggregator:

$$Y_o = A_o \left(\sum_{t \in T_o} \omega_{ot}^{\frac{1}{\eta}} y_{ot}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}. \quad (\text{A.20})$$

Define total labor and automation services in occupation o :

$$L_o := \sum_{t \in T_o} \ell_{ot}, \quad K_o := \sum_{t \in T_o} k_{ot},$$

and define the task-weighted automation capability (occupation exposure) as

$$\mathcal{I}_o := \sum_{t \in T_o} \omega_{ot} I_{ot} \in [0, 1]. \quad (\text{A.21})$$

A.3.2 Occupation-Level Reduced Form and Unit Cost

Under cost-minimizing allocation across tasks for given totals (L_o, K_o) , the nested CES structure (A.19)–(A.20) reduces to an occupation-level CES between effective labor and automation services:

$$Y_o = A_o \left((1 - \mathcal{I}_o)^{\frac{1}{\eta}} (e^{\pi_o} L_o)^{\frac{\eta-1}{\eta}} + (\mathcal{I}_o)^{\frac{1}{\eta}} (K_o)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}. \quad (\text{A.22})$$

Producers buy automation services at a fixed price $p_k > 0$ (perfectly elastic supply). Given (w_o, p_k) , cost minimization implies the standard CES input ratio in efficiency units:

$$\frac{K_o}{e^{\pi_o} L_o} = \frac{\mathcal{I}_o}{1 - \mathcal{I}_o} \left(\frac{w_o}{p_k e^{\pi_o}} \right)^{\eta}. \quad (\text{A.23})$$

The corresponding unit cost (and competitive output price) is

$$p_o = c_o(w_o, p_k; \pi_o, \mathcal{I}_o) = \frac{1}{A_o} \left[(1 - \mathcal{I}_o) \left(\frac{w_o}{e^{\pi_o}} \right)^{1-\eta} + \mathcal{I}_o p_k^{1-\eta} \right]^{\frac{1}{1-\eta}}. \quad (\text{A.24})$$

A.3.3 Labor Demand with Automation Services

Final demand remains CES across occupations as in (3). Under perfect competition, $p_o = c_o(\cdot)$ and CES demand implies $Y_o = \vartheta_o p_o^{-\sigma} Y$. Combining demand with (A.24) and the implied factor shares yields an occupational labor-demand schedule:

$$L_o^D(w_o; \tilde{\Theta}_o) = D_o e^{(\sigma-1)\pi_o} w_o^{-\sigma} (1 - \mathcal{I}_o) \left[(1 - \mathcal{I}_o) + \mathcal{I}_o \left(\frac{p_k e^{\pi_o}}{w_o} \right)^{1-\eta} \right]^{\frac{\sigma-\eta}{\eta-1}}, \quad (\text{A.25})$$

where $D_o := Y \vartheta_o A_o^{\sigma-1}$ is an occupation-specific shifter and $\tilde{\Theta}_o = (p_k, \pi_o, \mathcal{I}_o, \eta, \sigma)$.

A.3.4 Interpretation and Relation to the Benchmark Model

The benchmark model corresponds to a limit case of this extension. If $\mathcal{I}_o = 0$ for all occupations (no scope for automation services), or if p_k is sufficiently large that the cost-minimizing mix uses essentially no automation services, then (A.25) collapses to the familiar CES demand schedule $L_o^D = D_o e^{(\sigma-1)\pi_o} w_o^{-\sigma}$, and the only channels shaping wages are productivity and scarcity.

In the main text, we focus on the productivity–scarcity race using the benchmark model, and we treat automation and displacement as an additional demand-side channel whose incidence we explore only through sensitivity analysis (Appendix A.4).

A.4 Sensitivity of the Automation Channel

This appendix shows that the distributional effect of the automation channel depends on parameters that available data do not pin down. For average wages, the sign may be more constrained: Autor and Kausik (2026) show that when the labor share exceeds its wage-maximizing level—a condition they estimate holds across twelve industrialized

economies—automation that further lowers the labor share raises average wages. Its effect on inequality, the object of this appendix, is not pinned down.

A.4.1 Additional Parameters and the Sign of the Automation Channel

Relative to the benchmark model, the extension needs three objects that the data do not provide. The first is the effective price of automation services, p_k . This is not the posted price of a model but the full cost of using automation in production, including integration, supervision, compliance and liability, and it is not observed at the occupation level. The second is the within-occupation elasticity of substitution between labor and automation services, η , which governs how strongly firms substitute toward automation when relative prices change (Equation (A.23)). The third is the mapping from task exposure to automation capability $\mathcal{I}_o$: exposure measures say which tasks can be automated, not how much automation firms adopt, which depends on p_k and η . The final-demand elasticity σ also matters, because it governs how much output expands when unit costs fall.

These parameters matter because more automation capability has two opposing effects on labor demand. Holding output fixed, it shifts the input mix away from labor (a substitution effect). By lowering unit costs, it raises output demand and with it the demand for all inputs (a scale effect). Let $u_o := (w_o / (p_k e^{\pi_o}))^{\eta-1}$ and $Q_o := (1 - \mathcal{I}_o) + \mathcal{I}_o u_o$. Holding wages fixed, Equation (A.25) implies

$$\left. \frac{\partial \log L_o^D}{\partial \mathcal{I}_o} \right|_w = -\frac{1}{1 - \mathcal{I}_o} + \frac{\sigma - \eta}{\eta - 1} \cdot \frac{u_o - 1}{Q_o}. \quad (\text{A.26})$$

The first term is the substitution effect. The second is the scale effect, whose sign and size depend on $\sigma - \eta$ and on the relative price of automation services. When $\sigma < \eta$, substitution tends to dominate and labor demand falls; when $\sigma > \eta$, the scale effect can offset or reverse it.

The same forces determine which occupations are affected. For a given p_k , the relative price of automation services, $p_k e^{\pi_o} / w_o$, tends to be lower in high-wage occupations, and these occupations also have more automatable tasks (an employment-weighted automatable share of 32 percent in the top two quintiles of occupational wages, against 22 percent in the bottom quintile). Where substitution dominates, automation therefore compresses

wages at the top; where the scale effect is strong, the same occupations can gain. Because wages adjust in general equilibrium, these forces interact, and the incidence of automation is especially sensitive to (p_k, η, σ) .

A.4.2 Quantitative Sensitivity

We implement the extension of Appendix A.3 on the same data and calibration as the benchmark model, with $\mathcal{I}_o$ equal to the task-weighted share of the occupation's tasks that GenAI can automate (Section 4.1.2). With $\mathcal{I}_o = 0$ it reproduces the benchmark model exactly. We recompute the automation-only counterfactual in general equilibrium for the values in Table A.2, with p_k expressed as a multiple of the median occupational wage.

Table A.2: Automation Sensitivity Grid: Parameter Values

Parameter	Values
Price of automation services p_k (multiple of the median occupational wage)	0.25, 0.5, 0.75, 1, 1.5, 2
Within-occupation elasticity of substitution η	2, 4, 6, 10, 14, 20
Elasticity of substitution across occupations σ	4; also 2 and 8

Notes: The values of p_k and η are illustrative, not calibrated. Figure A.1 reports all 36 combinations of p_k and η at $\sigma = 4$. Table A.3 reports selected combinations, including $\sigma = 2$ and $\sigma = 8$ at $p_k = 1$ and $\eta = 10$.

Table A.3: Wage Inequality Under the Automation-Only Counterfactual: Sensitivity

Calibration	Var(log w)	$p50 - p10$	$p90 - p50$	$p90 - p10$
No automation	0.164	0.348	0.658	1.006
$p_k = 0.25, \eta = 20$	0.150	0.129	0.731	0.860
$p_k = 1, \eta = 10$	0.109	0.318	0.487	0.804
$p_k = 2, \eta = 2$	0.177	0.326	0.590	0.916
$p_k = 1, \eta = 10, \sigma = 2$	0.094	0.308	0.448	0.756
$p_k = 1, \eta = 10, \sigma = 8$	0.140	0.332	0.578	0.910

Notes: Moments of log occupational wages in general equilibrium, weighted by the model's employment in each scenario, as in Table 3. The first row is the economy without automation (Table 3). Rows two to four set $\sigma = 4$. p_k is relative to the median occupational wage.

The effect of the automation channel is not pinned down. Across the grid, it ranges from a 66 percent fall to more than a doubling of the variance of log wages, and its sign

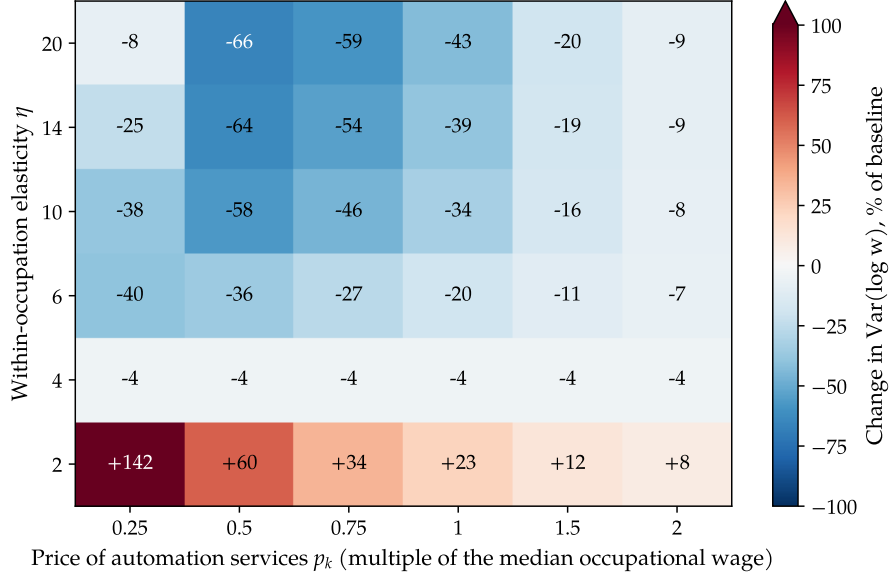


Figure A.1: Effect of the Automation Channel on Wage Inequality, by (p_k, η)

Notes: Each cell reports the general-equilibrium change in the variance of log wages under the automation-only counterfactual, in percent of the baseline variance (0.164), with $\sigma = 4$. In the row $\eta = 4$, $\eta = \sigma$, so the scale effect in Equation (A.26) is zero and the effect does not vary with p_k . Red cells widen and blue cells compress the wage distribution; the color scale is capped at ± 100 percent. Variances are weighted by the model's employment in each scenario, as in Table 3.

depends on η relative to σ , consistent with (A.26): automation tends to widen the wage distribution when η is below σ and compresses it when η is above σ (Figure A.1 and Table A.3). Where it compresses, it does so by lowering wages most at the top of the distribution (Figure A.2). When η is 10 or higher, the effect is not monotone in p_k : the compression is largest at $p_k = 0.5$ and smaller with cheaper automation (at $\eta = 20$, -66 percent at $p_k = 0.5$ against -8 percent at $p_k = 0.25$). With very cheap automation, wages fall in the middle of the distribution as well as at the top: at $p_k = 0.25$ and $\eta = 20$, the $p_{50} - p_{10}$ gap falls from 0.35 to 0.13, while the $p_{90} - p_{50}$ gap widens from 0.66 to 0.73 (Table A.3).

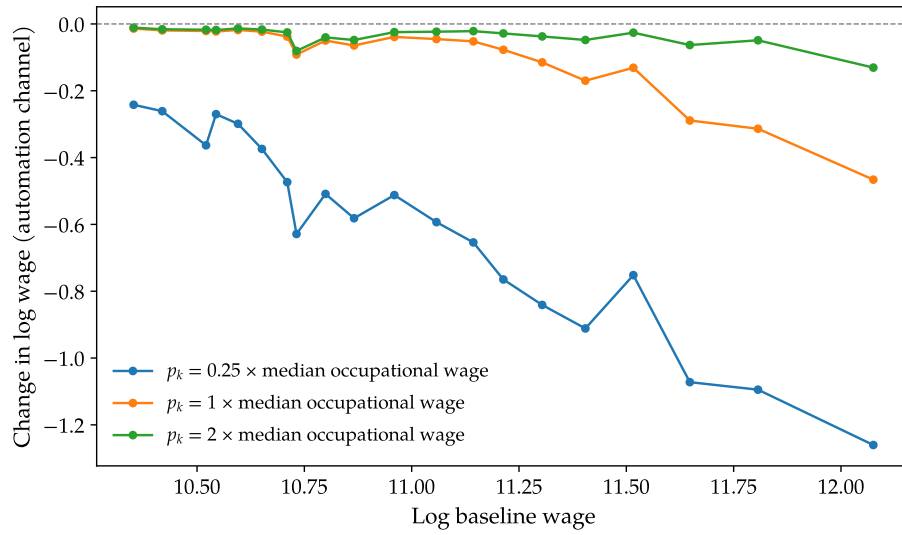


Figure A.2: Automation-Only Wage Changes by Baseline Wage

Notes: Employment-weighted mean change in log occupational wages within 20 bins of baseline log wages (bins of equal observed employment), under the automation-only counterfactual in general equilibrium, for three values of p_k ; $\eta = 10$ and $\sigma = 4$.

B Measurement

B.1 LLM Prompts

This appendix reproduces the prompts used to construct the LLM-based measures, verbatim from the code, including typos. Each prompt is sent as the system message; the user message, shown below it, supplies the task or the pair of occupations, which appear here as placeholders in braces. Table B.1 summarizes the five prompts. Three of them contain the same list of what an LLM can and cannot do; we print it once, in Appendix B.1.3, and omit it from the other two.

Table B.1: LLM Prompts

Measure	Model	Input	Output used	Section
Baseline expertise (1–5)	GPT-5.2	Task	expertise_score	4.1.1
Training time (months)	GPT-5.2	Task	training_months	4.1.1
Automation exposure (T0–T4)	GPT-5.2	Occupation and task	T_today	4.1.2
Expertise with GenAI (1–5)	GPT-5.2	Task	expertise_with_llm	4.1
Retraining distance (months)	GPT-5 Nano	Pair of occupations	retraining_months	6.2

Notes: The first four prompts are applied to each of the 19,265 O*NET tasks; the automation prompt also receives the task’s occupation. The retraining prompt is applied to each ordered pair of detailed occupations, once without and once with GenAI assistance.

B.1.1 Baseline Expertise

This prompt rates the expertise required for each task on a 1–5 scale (Section 4.1.1). For the validation in Appendix B.2, the same prompt is sent to Llama 3.3 70B and Claude Haiku 4.5.

System message:

You are rating the expertise required to perform individual job tasks from O*NET.

By "expertise" we mean specialized knowledge, training, or skill that:

- acts as a barrier to entry (not everyone can do it), and
- typically commands a wage premium.

Generic physical or social tasks that almost any adult could do with minimal training are "no or low expertise".

Give each task a 1-5 expertise score:

- 1 = No or minimal expertise - Generic or very basic tasks - Can be learned quickly with little training
- 2 = Low expertise - Some short training required - Limited occupation-specific knowledge
- 3 = Moderate expertise - Solid occupation-specific knowledge - Often some credential, apprenticeship, or substantial on-the-job learning
- 4 = High expertise - Advanced specialized knowledge - Significant training, degree, or certification
- 5 = Very high expertise - Deep specialized expertise - Often advanced professional or graduate-level training

Output STRICT JSON ONLY:

```
{
  "expertise_score": 1-5 integer,
  "expertise_label": "no_or_minimal" | "low" | "moderate" | "high" | "very_high",
  "confidence": 0.0-1.0
}
```

Do NOT include explanations.

Do NOT include text before or after the JSON.

Output ONLY valid JSON. No markdown. No comments. No words before or after.

User message:

Task:
{task}

Rate the expertise required.

B.1.2 Training Time

This prompt estimates the training time required for each task, which we use to map expertise categories into months (Section 4.1.1).

System message:

You are rating the expertise required to perform individual job tasks from O*NET.

For each task, estimate the minimum amount of specific education and training time (in months) required for a typical adult to learn to perform this task at a professional level.

Guidelines:

- Consider specialized knowledge, training, or skill needed.

- Focus on training/education time required to reach professional competence, not years of generic work experience.
- Respond with a single number in months (e.g., 0.5, 6, 18, 36, 60).

Output STRICT JSON ONLY:

```
{
  "training_months": number,
  "confidence": 0.0-1.0
}
```

Do NOT include explanations.

Do NOT include text before or after the JSON.

Output ONLY valid JSON. No markdown. No comments. No words before or after.

User message:

Task:
{task}

Estimate the minimum specific education and training time (in months) required for a typical adult to perform this task at a professional level.

B.1.3 Automation Exposure

This prompt classifies each task by the share of its components that current GenAI can perform (Section 4.1.2). Categories T0 to T4 correspond to categories 1 to 5 in the main text, and tasks rated T3 or T4 are classified as automatable ($\alpha_{ot} = 1$).

System message:

ROLE & PERSONA

You are a "Dual-Expert" evaluator with two distinct, high-level competencies:

1. ****Domain Expert:****

You deeply understand the occupation listed below, including tacit knowledge, informal practices, interpersonal nuance, real-world context, and the hidden complexity that non-experts overlook.

2. ****Gen AI Expert:****

You understand the real capabilities and limitations of modern LLMs, agentic workflows, RAG systems, tool use (Python/APIs), and multimodal models. You can distinguish between "demo-level" and "production-reliable" AI behavior.

Your job is to integrate BOTH perspectives.

```

### GOAL
Evaluate the automation exposure of a specific occupational task:
  **Cuurent December 2025 LLM abilities

---

### AUTOMATION RUBRIC (T0-T4 -- USE EXACT DEFINITIONS)

**T0 -- No Automation**
System cannot perform any meaningful component of the task. Typically highly physical,
  deeply emotional, or restricted by legal requirements.

**T1 -- Low Automation**
System can perform 0-50% of the task. Core work relies on physical action, real-world
  perception, in-person nuance, or tacit knowledge AI cannot substitute.

**T2 -- Moderate Automation (Hybrid)**
System can perform 50-80% of the task at high quality.
Human involvement remains essential for:
- physical actions,
- in-person communication,
- or tacit, context-dependent judgment.

**T3 -- High Automation (Human-in-the-Loop)**
System can perform 80-100% at high quality, BUT human oversight is required because of:
- liability/safety,
- stakeholder trust expectations,
- rare catastrophic failure modes.

**T4 -- Full Automation (Autonomous)**
System performs 100% of the task with high quality.
Human oversight is *not routinely needed* and humans are *not liable* for errors.
The entire workflow is digitally executable end-to-end.

---

### CAPABILITY ASSUMPTIONS

#### LLM(Current, Year 2025)**
**CAN:**
- Multi-step planning and reasoning
- Advanced multimodal analysis (vision/audio/docs/charts)
- Retrieval-augmented generation with large knowledge bases
- API/Python/Excel/SQL tool use
- Code generation, debugging, and software automation
- Structured workflow execution and agentic autonomy
- Up-to-date factual retrieval via internet tools
- Data analysis, visualization, and cleaning
- Idea generation, brainstorming, and research assistance
- High-quality writing, summarization, and translation
- Professional communication and long-context coherence

```

- Document extraction, comparison, and synthesis
- Business analytics, forecasting, and competitive intelligence
- UX/UI design, creative writing, images, audio, and video generation
- App/website creation and end-to-end automation
- Long-term memory and personalization (where enabled)
- Self-correction, critique, and verification loops
- Persistent agentic behaviors across sessions
- Scientific reasoning: literature synthesis, experimental design

****CANNOT:****

- Perform any physical actions
- Conduct tactile or in-person inspection
- Guarantee correctness in high-stakes situations
- Substitute for human judgment where regulation requires a human
- Provide genuine emotional presence or tacit human understanding

OUTPUT FORMAT (MANDATORY JSON)

Return exactly:

```
{
  "T_today": "T0 | T1 | T2 | T3 | T4"
}
```

User message:

Occupation: {occupation title}

Task Description: {task}

Using the rubric and capability assumptions above, evaluate the automation exposure for this task and respond with the required JSON object.

B.1.4 Expertise with GenAI Assistance

This prompt measures the intensive margin (Section 4.1). It elicits four values, of which we use only `expertise_with_llm`, the 1–5 rating with a GenAI assistant, which we map into training months with the occupation-specific mapping. The baseline rating comes from the prompt in Appendix B.1.1.

System message:

ROLE & PERSONA

You are a "Dual-Expert" evaluator with two distinct, high-level competencies:

1. ****Domain Expert:****

You deeply understand the occupation listed below, including tacit knowledge,

informal practices, interpersonal nuance, real-world context, and the hidden complexity that non-experts overlook.

2. ****Gen AI Expert:****

You understand the real capabilities and limitations of modern LLMs, agentic workflows, RAG systems, tool use (Python/APIs), and multimodal models. You can distinguish between "demo-level" and "production-reliable" AI behavior.

Your job is to integrate BOTH perspectives.

GOAL

For the task below, provide TWO expertise ratings AND TWO training-time estimates:

- 1) WITHOUT access to an LLM assistant
- 2) WITH access to a capable LLM assistant (current December 2025 abilities)

Expertise means specialized knowledge, training, or skill that:

- acts as a barrier to entry (not everyone can do it), and
- typically commands a wage premium.

Use the SAME 1-5 scale for both expertise ratings:

- 1 = No or minimal expertise - Generic/basic tasks - learn quickly with little training
- 2 = Low expertise - short training - limited occupation-specific knowledge
- 3 = Moderate expertise - solid occupation-specific knowledge - often credential/apprenticeship/substantial on-the-job learning
- 4 = High expertise - advanced specialized knowledge - significant training/degree/certification
- 5 = Very high expertise - deep specialized expertise - often advanced professional/graduate-level training

Training-time estimate:

- Report the MINIMUM specific education/training time (in months) required for a typical adult to learn to perform the task at a professional level.
- Focus on task-specific training/education time, not generic years of work experience.
- Months may be fractional (e.g., 0.5, 6, 18, 36, 60).

CAPABILITY ASSUMPTIONS (ONLY FOR THE "WITH LLM" RATING)

[List of what the LLM can and cannot do, identical to the list in Appendix B.1.3.]

OUTPUT FORMAT (MANDATORY JSON)

Return exactly:

```
{
  "expertise_without_llm": 1-5 integer,
  "training_months_without_llm": number,
  "expertise_with_llm": 1-5 integer,
  "training_months_with_llm": number
}
```

No other keys.

No explanations.
Output ONLY valid JSON. No markdown. No text before or after.

User message:

Task Description: {task}

Using the rubric and capability assumptions above, return the required JSON object with the two expertise ratings and two training-month estimates.

B.1.5 Retraining Distances

These prompts estimate the retraining distances of the dynamic model (Section 6.2), using GPT-5 Nano. Each request covers one ordered pair of detailed occupations. The user message lists each occupation's title and up to 30 of its core tasks, each truncated at 200 characters, and is the same in both scenarios; only the system message differs.

System message, without GenAI:

You estimate months of RETRAINING needed for an average worker in an ORIGIN occupation to become able to perform a TARGET occupation at a professional level.

Definitions:

- "Average worker in ORIGIN" = someone currently competent at the listed ORIGIN tasks, with typical occupation-specific knowledge/credentials implied by the occupation.
- "Retraining months" = additional targeted education/training beyond what the worker already has from ORIGIN (courses, certification prep, supervised practice).
- Do NOT count generic work experience time. Count focused learning time.
- Assume full-time retraining (40 hrs/week). Months can be fractional.

Return ONLY JSON with keys:

```
{  
  "retraining_months": number,  
  "confidence": 0.0-1.0  
}
```

No extra text.

System message, with GenAI:

You estimate months of RETRAINING needed for an average worker in an ORIGIN occupation to become able to perform a TARGET occupation at a professional level, UNDER THE ASSUMPTION THAT THE WORKER HAS ACCESS TO A POWERFUL LLM ASSISTANT (Year 2025 capabilities).

Definitions:

- "Average worker in ORIGIN" = someone currently competent at the listed ORIGIN tasks, with typical occupation-specific knowledge/credentials implied by the occupation.
- "Retraining months" = additional targeted education/training beyond what the worker already has from ORIGIN (courses, certification prep, supervised practice).
- Do NOT count generic work experience time. Count focused learning time.
- Assume full-time retraining (40 hrs/week). Months can be fractional.

IMPORTANT:

The LLM assistant does NOT replace the worker.

It reduces learning time for knowledge-intensive, analytical, documentation, coding, research, planning, writing, and information-processing components.

It does NOT reduce time required for:

- Physical skills
- Manual dexterity
- In-person supervision requirements
- Tacit experiential judgment that must be practiced physically

CAPABILITY ASSUMPTIONS

[List of what the LLM can and cannot do, identical to the list in Appendix B.1.3.]

Return ONLY JSON with keys:

```
{
  "retraining_months": number,
  "confidence": 0.0-1.0
}
```

No extra text.

User message (both scenarios):

```
ORIGIN occupation: {origin title}
ORIGIN typical tasks:
- {task}
- {task}
...
```

```
TARGET occupation: {target title}
TARGET typical tasks:
- {task}
- {task}
...
```

Return JSON only.

B.2 Validation of the LLM-Based Expertise Measure

Figure B.1 compares the occupation-level expertise requirements computed from GPT-5.2 ratings with those computed from the ratings of Llama 3.3 70B and Claude Haiku 4.5, which received the same prompt (Appendix B.1.1). The R^2 is 0.96 in both cases, so the measure is stable across models.

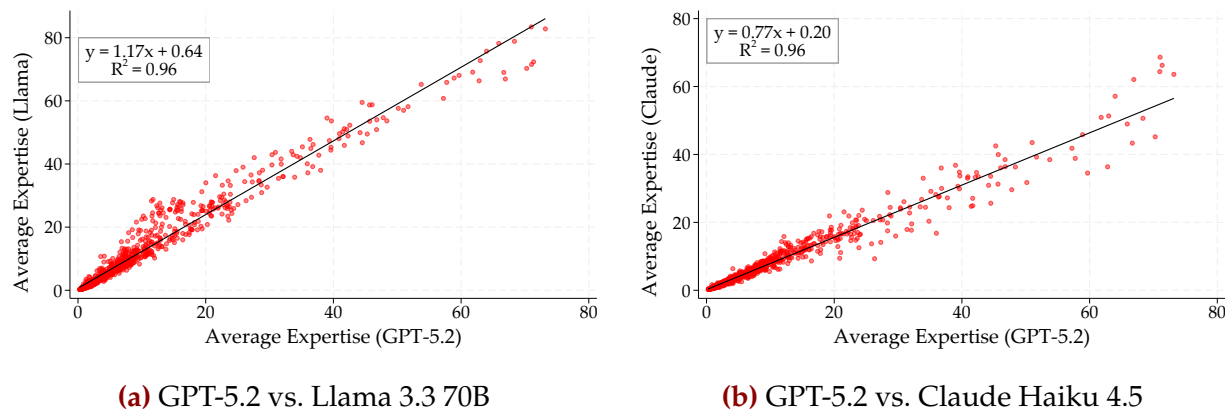


Figure B.1: Occupational Expertise Requirements Across LLMs

Notes: Each point is an occupation. The figure compares occupation-level expertise requirements computed from GPT-5.2 ratings with those computed from Llama 3.3 70B ratings (Panel a) and Claude Haiku 4.5 ratings (Panel b). The line is an OLS fit.

We also merge in occupation characteristics from the O*NET 30.0 Database, including the levels of education and experience required for each occupation. These data are collected through questionnaires that ask job incumbents to report the education and experience expected of a new hire.³⁴ We additionally merge average and median wage levels from the May 2024 Occupational Employment and Wage Statistics (OEWS) provided by the BLS.³⁵ Figure B.2 shows that occupations with higher expertise requirements require more education ($R^2 = 0.61$) and pay higher mean and median wages ($R^2 = 0.52$ and 0.40). The relationship with required experience is also positive but weaker ($R^2 = 0.12$).

³⁴For the O*NET 30.0 Education, Training, and Experience Database, see onetcntr.org/dictionary/30.0/excel. For the questionnaires, see onetcntr.org/questionnaires. The dataset reports the share of incumbents selecting each education and experience category. We convert these categories into years and compute weighted averages.

³⁵For the OEWS data, see <https://www.bls.gov/oes>.

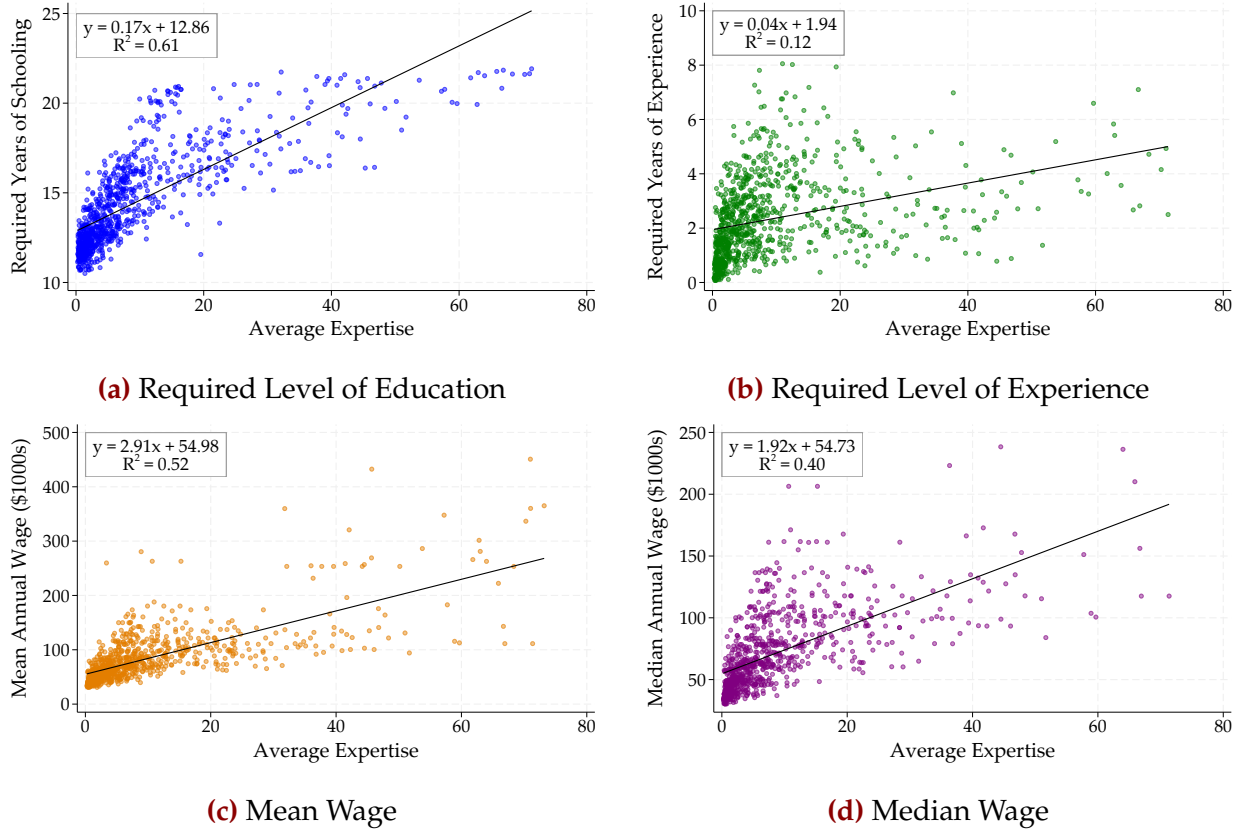


Figure B.2: Occupational Expertise Requirements and Occupational Characteristics

Notes: Each point is an occupation. The figure plots the occupation-level expertise requirement against required years of education (Panel a), required years of experience (Panel b), and the mean and median annual wage in thousands of dollars (Panels c and d). The line is an OLS fit.

B.3 Distributions of Expertise, PSS and PPG

Figure B.3 shows the distribution of task-level expertise ratings with and without GenAI assistance, and Figure B.5 shows the distributions of PSS and PPG across occupations. Figure B.4 plots the distributions of four occupation-level expertise measures, each computed as a task-weighted average (with weights of 1 for core tasks and 0.5 for supplemental tasks) under different GenAI scenarios. The *baseline* measure X_0^{noAI} averages the baseline training months of the occupation's tasks, x_{ot}^{noAI} (Equation (18)). The *extensive-margin* measure $X_0^{extensive}$ averages expertise over only the tasks that are *not* classified as automatable (tasks with $\alpha_{ot} = 0$), thereby capturing the effect of removing automated tasks from the occupation's task bundle. The *intensive-margin* measure $X_0^{intensive}$ averages the training months with GenAI assistance, x_{ot}^{withAI} , over the full set of tasks, capturing

the reduction in required expertise when workers perform all tasks with GenAI assistance. Finally, the *combined* measure $\chi_o^{combined}$ applies both margins simultaneously: it averages training months with GenAI assistance over the non-automated tasks only, as defined in Equation (19).

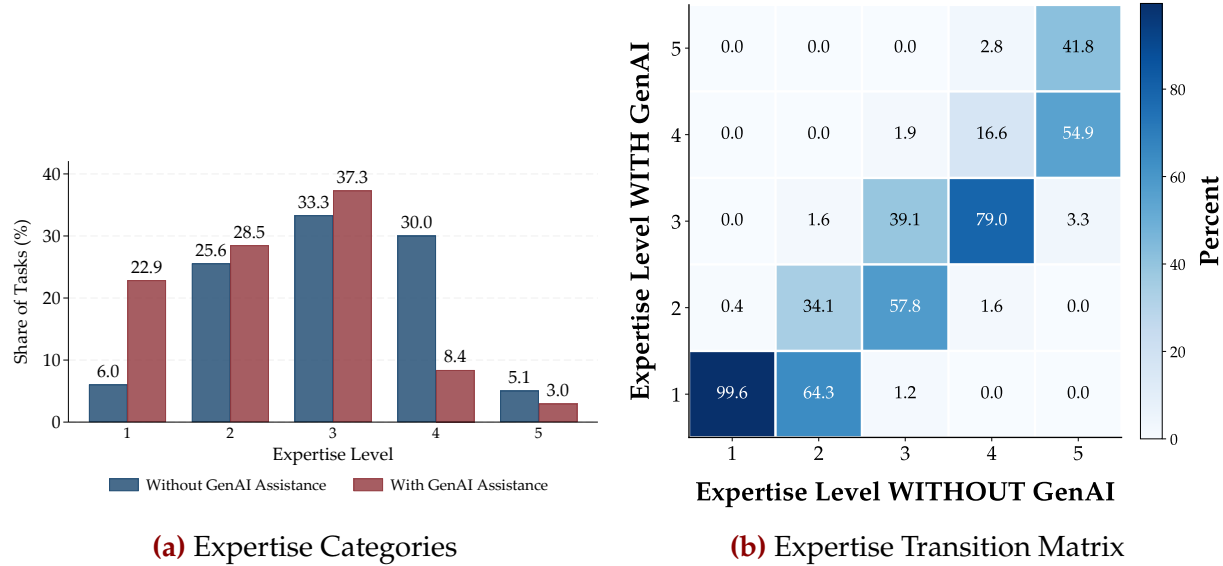


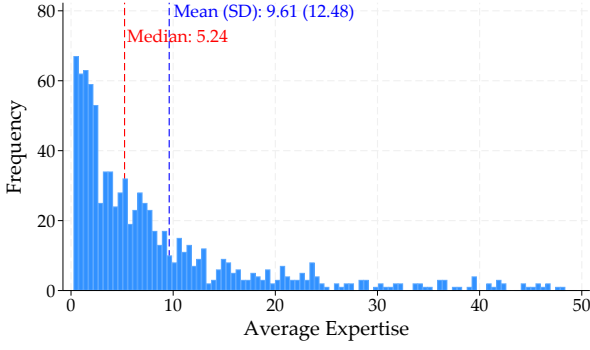
Figure B.3: The Expertise Distribution of Tasks With and Without GenAI Assistance

Notes: Panel (a) shows the distribution of tasks across the five expertise categories without (blue) and with (red) GenAI assistance. Panel (b) shows transitions between categories: each cell gives the percentage of tasks in a category without GenAI (x-axis) that are in a given category with GenAI (y-axis).

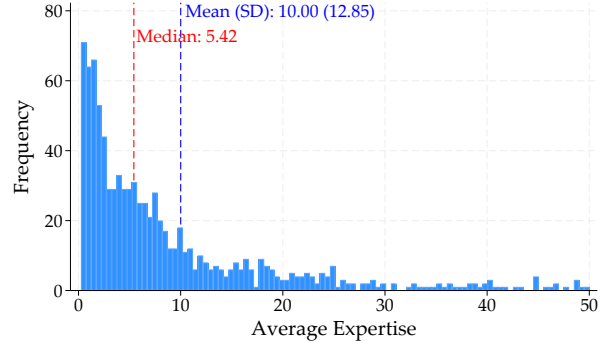
B.4 Occupations with the Highest and Lowest Expertise and PSS

Tables B.2 and B.3 list the twenty occupations at each end of the distributions of baseline expertise requirements χ_o^{noAI} and of the Potential Supply Shift, with the baseline and post-GenAI requirements in months, the automatable share of tasks, PSS, and 2024 employment.

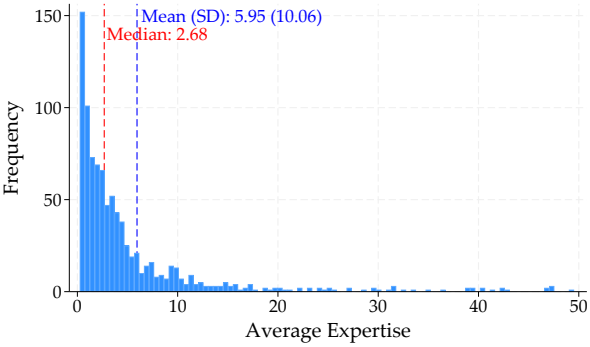
The expertise ranking in Table B.2 is as one would expect. The highest requirements belong to surgeons, physicians, judges, lawyers and research scientists, at four to six years of training beyond a typical adult's starting point; the lowest belong to dishwashers, cleaners, food servers, packers and attendants, at one to two weeks. The top of the ranking also illustrates why PSS is small for the most expert occupations: GenAI lowers



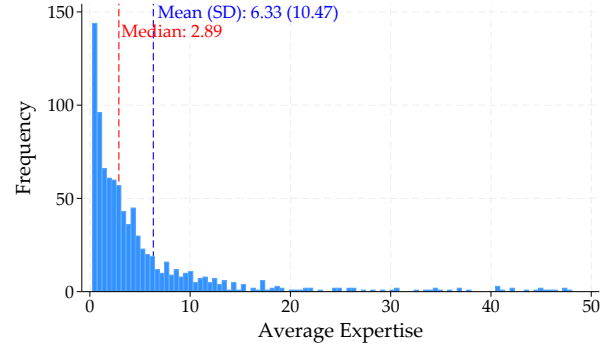
(a) Without GenAI Effects



(b) Extensive



(c) Intensive



(d) Combined

Figure B.4: Average Expertise Across Occupations

Notes: Each panel plots the distribution of occupation-level expertise requirements, measured in mapped training months. Panel (a): baseline expertise χ_o^{noAI} . Panel (b): extensive-margin expertise $\chi_o^{extensive}$ (averaging over non-automated tasks only). Panel (c): intensive-margin expertise $\chi_o^{intensive}$ (all tasks re-rated with GenAI assistance). Panel (d): combined expertise $\chi_o^{combined}$ (GenAI-assisted training months over non-automated tasks).

their requirements by months or even years in absolute terms (for example, from 60 to 34 months for Medical Scientists), but because so few workers have expertise anywhere near the new threshold, the share of the workforce newly qualified is close to zero.

Table B.3 shows the occupations in which GenAI expands the eligible pool the most and the least. The largest PSS values (0.30 to 0.54) arise where a low or moderate baseline requirement, between roughly two and eighteen months, falls to a few months or less. In these occupations the intensive margin re-rates most tasks one category down, and the drop in the requirement spans the dense part of the expertise distribution, where a

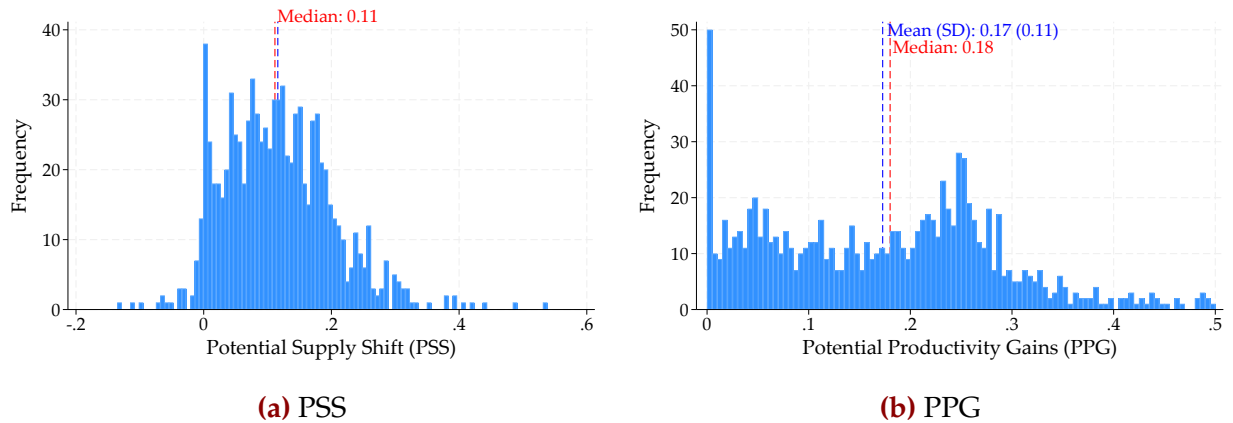


Figure B.5: The Distributions of PSS and PPG Across Occupations

Notes: Panel (a) shows the distribution of the Potential Supply Shift, $F(X_0^{noAI}) - F(X_0^{combined})$. Panel (b) shows the distribution of Potential Productivity Gains, defined in Equation (21).

large share of the workforce is located. Pharmacy Aides, Credit Analysts, Chemists, Industrial Engineers and Data Scientists are examples. The bottom of the table consists of occupations with negative PSS, that is, occupations whose requirement *rises* with GenAI. This happens for two reasons. First, the extensive margin removes tasks, and when the automated tasks are the easier ones the average requirement of the remaining bundle rises (Customer Service Representatives, with 86 percent of tasks automatable, and Proofreaders, with 90 percent). Second, the LLM rates some tasks as requiring *more* expertise with GenAI assistance, often hands-on or person-facing tasks (Emergency Medical Technicians, Lifeguards, Dental Hygienists); since the baseline and with-GenAI ratings come from different prompts, some of these increases are likely noise. Negative PSS is uncommon: it applies to 41 occupations (4.6 percent of occupations, 4.8 percent of employment), and we keep it as measured rather than censoring it, since the model allows barriers to rise as well as fall.

Table B.2: Occupations with the Highest and Lowest Baseline Expertise Requirements

Occupation	R_o	$\tilde{R}_o$	PSS	Empl. (000s)
<i>Panel A: 20 highest baseline expertise requirements</i>				
Orthopedic Surgeons, Except Pediatric	73.2	75.3	+0.000	14.2
Clinical Neuropsychologists	71.4	63.5	+0.002	8.9
Oral and Maxillofacial Surgeons	71.0	61.1	+0.002	5.3
Pediatric Surgeons	71.0	67.4	+0.001	1.1
Anesthesiologists	70.3	68.4	+0.000	41.9
Urologists	68.4	76.9	-0.001	52.6
Neuropsychologists	66.9	57.2	+0.008	8.9
Judges, Magistrate Judges, and Magistrates	66.7	63.4	+0.001	25.6
Pediatricians, General	65.9	58.5	+0.002	43.0
General Internal Medicine Physicians	64.0	52.0	+0.007	66.6
Obstetricians and Gynecologists	63.0	56.5	+0.007	19.9
Ophthalmologists, Except Pediatric	62.8	55.6	+0.007	12.1
Physicians, Pathologists	61.8	56.1	+0.007	11.8
Medical Scientists, Except Epidemiologists	59.7	33.6	+0.023	156.3
Biochemists and Biophysicists	58.9	42.2	+0.009	34.5
Lawyers	57.8	45.6	+0.007	747.8
Dermatologists	57.2	47.4	+0.001	10.1
Neurologists	53.8	47.5	+0.001	7.7
Epidemiologists	51.7	24.2	+0.021	11.5
Economists	51.0	32.3	+0.016	7.9
<i>Panel B: 20 lowest baseline expertise requirements</i>				
Dishwashers	0.2	0.2	+0.000	471.7
Dining Room and Cafeteria Attendants and Bart...	0.3	0.3	+0.002	522.0
Maids and Housekeeping Cleaners	0.3	0.3	+0.009	854.9
Food Servers, Nonrestaurant	0.3	0.3	+0.010	271.8
Packers and Packagers, Hand	0.3	0.2	+0.016	601.4
Food Preparation Workers	0.3	0.3	+0.022	888.8
Couriers and Messengers	0.3	0.3	+0.020	71.9
Baristas	0.3	0.2	+0.036	1,890.5
Postal Service Mail Carriers	0.4	0.2	+0.038	336.0
Switchboard Operators, Including Answering Se...	0.4	0.2	+0.038	35.7
Ushers, Lobby Attendants, and Ticket Takers	0.4	0.3	+0.037	119.2
Graders and Sorters, Agricultural Products	0.4	0.3	+0.017	26.9
Locker Room, Coatroom, and Dressing Room Attendants	0.4	0.3	+0.036	15.0
Taxi Drivers	0.4	0.3	+0.038	17.5
Postal Service Mail Sorters, Processors, and...	0.4	0.3	+0.037	111.9
Helpers-Production Workers	0.4	0.3	+0.038	167.5
Baggage Porters and Bellhops	0.4	0.4	+0.002	31.2
Door-to-Door Sales Workers, News and Street V...	0.4	0.3	+0.034	4.6
Mail Clerks and Mail Machine Operators, Except...	0.4	0.3	+0.040	62.7
Parking Attendants	0.4	0.3	+0.041	134.7

Notes: R_o is the baseline occupation-level expertise requirement X_o^{noAI} and $\tilde{R}_o$ the post-GenAI requirement $X_o^{combined}$, both in months of training. $PSS = F(R_o) - F(\tilde{R}_o)$. Employment is 2024 BLS OEWS employment in thousands, split equally across O*NET occupations that share a six-digit SOC code. Long titles are truncated. Orthopedic Surgeons and Urologists are excluded from the counterfactuals (Section 5.1).

Table B.3: Occupations with the Highest and Lowest Potential Supply Shift

Occupation	R_o	$\tilde{R}_o$	Autom. share	PSS	Empl. (000s)
<i>Panel A: 20 highest PSS</i>					
Pharmacy Aides	4.0	0.4	0.48	+0.538	41.1
Desktop Publishers	3.3	0.4	0.94	+0.490	4.0
Credit Analysts	5.9	0.8	0.78	+0.440	67.4
Solar Energy Systems Engineers	16.8	2.7	0.40	+0.418	18.8
Facilities Managers	9.2	1.9	0.18	+0.405	70.5
Poets, Lyricists and Creative Writers	7.9	1.7	0.56	+0.395	23.9
Chemists	17.6	3.4	0.16	+0.394	83.2
Conservation Scientists	16.4	3.5	0.19	+0.382	8.5
Electrical and Electronic Engineering Technol. ...	12.1	2.6	0.37	+0.377	92.7
Industrial Engineers	12.7	3.3	0.39	+0.351	87.6
Multiple Machine Tool Setters, Operators, and ...	2.4	0.6	0.18	+0.331	129.8
Geothermal Production Managers	11.5	3.6	0.23	+0.327	39.1
Manufacturing Engineers	14.9	4.1	0.17	+0.321	87.6
Data Scientists	11.5	3.5	0.88	+0.318	77.8
Online Merchants	2.3	0.6	0.74	+0.316	282.1
Automotive Engineering Technicians	11.3	3.5	0.28	+0.316	18.7
Teaching Assistants, Postsecondary	6.8	2.1	0.30	+0.311	155.0
Compensation and Benefits Managers	14.3	4.1	0.34	+0.311	20.1
Legal Secretaries and Administrative Assistan. ...	2.8	0.7	0.61	+0.305	154.5
Shampooers	1.9	0.5	0.14	+0.305	8.9
<i>Panel B: 20 lowest PSS</i>					
Proofreaders and Copy Markers	1.7	3.0	0.90	−0.135	5.2
Emergency Medical Technicians	5.5	7.4	0.00	−0.113	178.0
Customer Service Representatives	2.1	3.7	0.86	−0.096	2,725.9
Lifeguards, Ski Patrol, and Other Recreationa. ...	1.7	2.2	0.00	−0.075	143.6
Helpers—Extraction Workers	2.0	2.3	0.00	−0.067	6.7
Correspondence Clerks	0.9	1.2	0.89	−0.066	6.3
Airline Pilots, Copilots, and Flight Engineers	8.9	11.4	0.43	−0.059	99.3
Bus Drivers, School	0.6	0.7	0.24	−0.050	387.9
Correctional Officers and Jailers	1.6	2.0	0.14	−0.039	365.4
Rail Car Repairers	3.9	4.4	0.00	−0.037	18.3
Broadcast Announcers and Radio Disc Jockeys	3.5	4.0	0.62	−0.035	23.9
Dental Hygienists	7.8	9.3	0.06	−0.032	219.1
Commercial Divers	6.6	7.3	0.00	−0.032	3.4
Bus Drivers, Transit and Intercity	0.5	0.6	0.26	−0.029	149.0
Commercial Pilots	11.0	12.2	0.36	−0.020	51.8
Surgical Assistants	7.2	7.7	0.00	−0.019	22.9
Customs and Border Protection Officers	15.1	15.5	0.10	−0.013	333.5
Low Vision Therapists, Orientation and Mobili. ...	9.6	10.4	0.05	−0.011	76.1
Subway and Streetcar Operators	0.8	0.9	0.27	−0.011	9.2
Animal Control Workers	6.3	6.6	0.13	−0.011	11.8

Notes: See Table B.2. Automatable share is the task-weighted share of the occupation's tasks in automation categories 4–5 (the extensive margin, Section 4.1.2). A negative PSS means that the requirement rises with GenAI, because automation removes easier tasks or because tasks are rated as requiring more expertise with GenAI assistance.

B.5 Comparing Automation Exposure Measures

This appendix compares our automation exposure measure with the original measure of [Eloundou et al. \(2024\)](#). For each occupation, Figure B.6 plots the share of tasks that are automatable under the original GPT-4-based measure, where a task counts as automatable if it is rated T3 or T4 on their automation rubric, against the share under our GPT-5.2-based measure, where a task counts as automatable if it is in category 4 or 5 (T3 or T4 in our prompt; Appendix B.1.3). The two measures are strongly related ($R^2 = 0.76$).

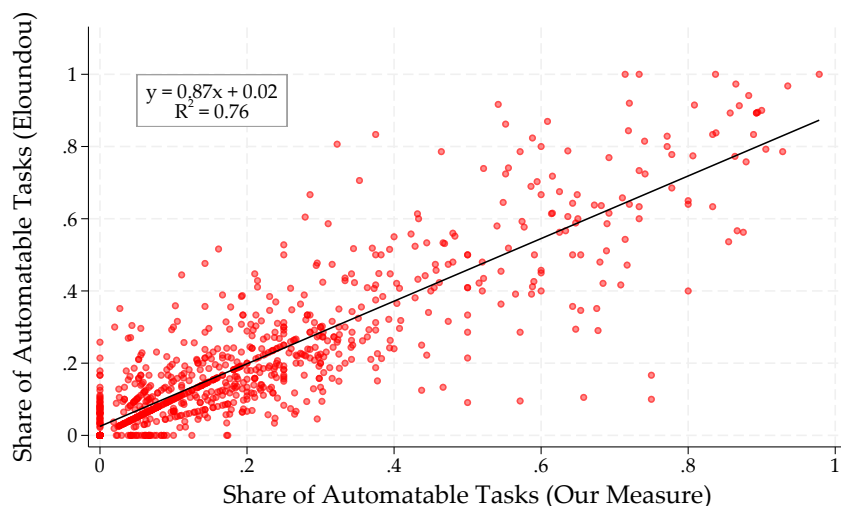


Figure B.6: Our Automation Exposure Measure and That of [Eloundou et al. \(2024\)](#)

Notes: Each point is an O*NET occupation. The vertical axis reports the share of the occupation's tasks that are automatable under the measure of [Eloundou et al. \(2024\)](#) and the horizontal axis the share under our GPT-5.2-based measure. The line is an OLS fit.

B.6 PSS Dispersion Among Occupations with Similar Exposure Shares

Figure B.7 reports PSS for the 32 occupations whose automatable share lies between 29 and 31 percent (Section 4.4).

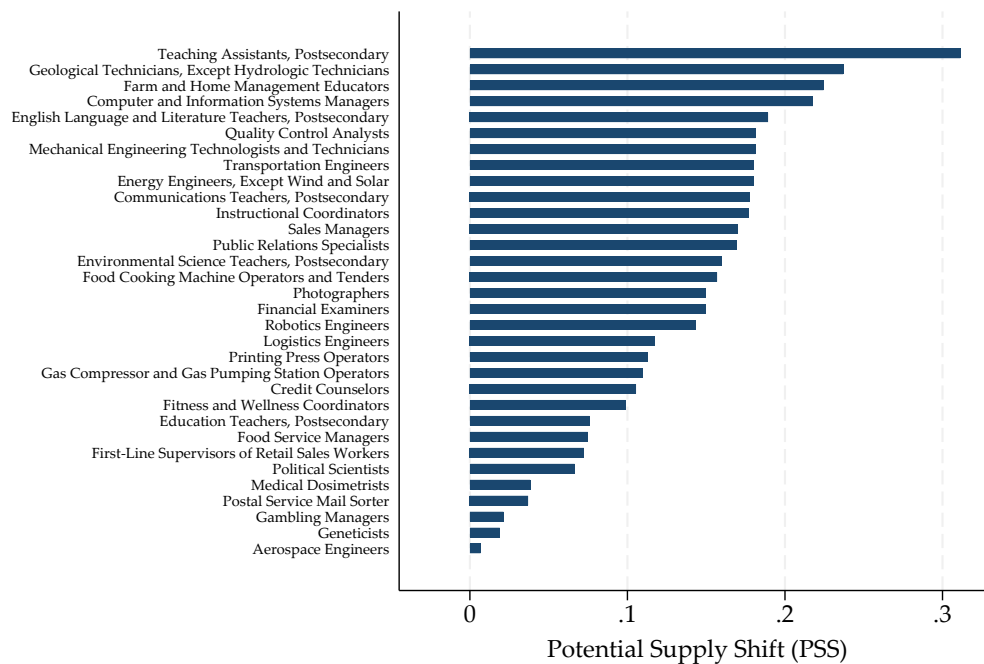


Figure B.7: PSS of Occupations with About 30 Percent of Tasks Automatable

Notes: PSS for all occupations with 29–31 percent of tasks automatable, around the 75th percentile of the automatable share across occupations.

C Static Model: Additional Results and Robustness

C.1 Decomposing the Change in Inequality

Let $V(w, L)$ denote the variance of log wages w across workers with employment weights L , and let subscripts 0 and 1 denote the baseline and a counterfactual. We split the change in inequality, $V(w_1, L_1) - V(w_0, L_0)$, into a *wage-structure effect*, $V(w_1, L_0) - V(w_0, L_0)$, which evaluates the new occupational wages at the baseline distribution of workers across occupations, and a *composition effect*, $V(w_1, L_1) - V(w_1, L_0)$, which captures the reallocation of workers across occupations at the new wages. Table C.1 reports both for the three counterfactuals of Section 5.2.

Table C.1: Decomposition of the Change in Inequality

	Total	Wage-structure effect	Composition effect
Supply channel	−8.3	−21.1	+12.7
Productivity channel	+24.9	+22.3	+2.6
Both channels	+13.1	−3.9	+16.9

Notes: Change in the employment-weighted variance of log occupational wages between the baseline and each general-equilibrium counterfactual, in percent of the baseline variance, at $\sigma = 4$. The split is sequential, with wages first; reversing the order gives wage-structure effects of −21.7, +23.6 and −1.9 percent and composition effects of +13.4, +1.3 and +14.9 percent.

Under the supply channel, the wage-structure effect is large: holding the allocation of workers fixed, the erosion of scarcity premia lowers the variance of log wages by 21 percent of its baseline value. The composition effect offsets most of it, adding back 13 percent, because workers move into higher-paid occupations. The net decline of 8 percent is therefore small not because the supply channel is weak, but because much of its effect takes the form of workers moving up rather than of wages at the top falling. Under the productivity channel, almost all of the change is a wage-structure effect, since feasibility does not change. With both channels, the wage-structure effect is slightly negative (−4 percent), so the net increase comes entirely from composition.

The two effects are not equally robust. The wage-structure effect is driven by the measured supply shift and the elasticity of substitution across occupations. The composition effect depends on how workers are allocated across occupations at baseline, which the model matches only loosely (Section 5.1), so its size, and with it the sign of the total

across workers, is more sensitive to the model’s structure. Under the education-based expertise distribution of Appendix C.4, the supply channel’s wage-structure effect remains negative but is about a third as large (−7 percent), and the composition effect is close to zero.

Table C.2 reports the same split for the supply channel at each value of σ in Table 4. The wage-structure effect is much larger when σ is low: −29 percent of baseline variance at $\sigma = 2$, against −21 percent at our baseline of $\sigma = 4$ and −13 percent at $\sigma = 8$. The composition effect barely moves, at +10 to +14 percent, because the supply shift itself does not depend on σ and wages only partly undo it. As σ rises, the total therefore shrinks, and at $\sigma = 8$ it turns slightly positive.

Table C.2: The Supply Channel by Elasticity of Substitution

σ	1.5	2	3	4	5	6	8
Total change	−21.5	−18.2	−12.6	−8.3	−5.1	−2.7	+0.9
Wage-structure effect	−31.2	−29.0	−24.6	−21.1	−18.3	−16.2	−13.1
Composition effect	+9.7	+10.8	+12.0	+12.7	+13.2	+13.5	+14.0

Notes: General-equilibrium change in the employment-weighted variance of log occupational wages under the supply channel, in percent of baseline variance, for each value of σ ; $\{B_o\}$ is recalibrated for each σ . The wage-structure and composition effects are defined as in Table C.1. The baseline is $\sigma = 4$.

C.2 Sensitivity of the Race to the Size of Productivity Gains

This appendix varies the size of productivity gains, which Section 5.3 holds fixed. PPG reads the exposure indicator of Eloundou et al. (2024) as a 50 percent time saving on directly exposed tasks and 25 percent on tasks exposed only with additional software (Section 4.2); these magnitudes are a calibration choice, and experimental estimates of GenAI time savings vary widely across tasks (Noy and Zhang, 2023; Dell’Acqua et al., 2023; Brynjolfsson et al., 2025b). Table C.3 scales PPG by one half, one and one and a half for each value of σ .³⁶ The productivity channel widens inequality in every case, and more than proportionally in the assumed time savings: at $\sigma = 4$ its effect goes from +10 to +25 to +53 percent. With half the baseline PPG, the threshold above which productivity wins

³⁶Doubling PPG is not feasible in the model, since occupations whose work volume is fully exposed would then have infinite productivity gains.

across workers rises from about $\sigma = 2.7$ to about $\sigma = 4.0$; with one and a half times the baseline PPG, it falls to about $\sigma = 2.0$. On the wage structure alone (middle block), the threshold is about $\sigma = 4.7$ with the baseline PPG, and scarcity wins at every σ in the table with half the baseline PPG.

Table C.3: Sensitivity of the Race to the Size of Productivity Gains

σ	Net effect, across workers			Net effect, wage structure			Productivity only		
	$\times 0.5$	$\times 1$	$\times 1.5$	$\times 0.5$	$\times 1$	$\times 1.5$	$\times 0.5$	$\times 1$	$\times 1.5$
2	-14.6	-9.4	-0.7	-26.2	-21.9	-15.1	+5.5	+13.2	+25.3
3	-6.2	+3.5	+21.3	-19.5	-11.7	+1.7	+8.1	+20.3	+41.7
4	0.0	+13.1	+38.0	-14.4	-3.9	+14.4	+9.7	+24.9	+53.2
5	+4.6	+20.2	+50.7	-10.5	+2.0	+24.0	+10.7	+28.0	+61.7
6	+8.1	+25.7	+60.6	-7.5	+6.5	+31.4	+11.5	+30.4	+68.2
8	+13.2	+33.5	+74.9	-3.2	+13.0	+42.0	+12.5	+33.6	+77.6

Notes: The table varies σ (rows) and the size of productivity gains (columns: PPG scaled by 0.5, 1 and 1.5 relative to the baseline). The first block reports the percentage change in the employment-weighted variance of log occupational wages between the baseline and the general-equilibrium counterfactual with both channels; the second block reports the wage-structure effect of the same counterfactual, as a percent of baseline variance; the third block reports the change in the variance for the productivity channel alone. $\{B_o\}$ is recalibrated for each σ ; the supply channel alone does not depend on PPG and is reported by σ in Table 4.

C.3 Sensitivity to the Aggregation Exponent ρ

The baseline analysis sets $\rho = 1$ in Equation (1), so that an occupation's entry barrier is the importance-weighted arithmetic mean of its task-level expertise requirements. This appendix examines the sensitivity of both descriptive and general-equilibrium results to the choice of ρ , considering values $\rho \in \{1, 2, 3, 5, 10\}$ that span from the arithmetic mean to a near-maximum aggregator.

C.3.1 Occupation-Level Expertise Rankings

Because the power mean is monotonically increasing in ρ , raising the aggregation exponent shifts the level of every occupation's entry barrier upward. The economically relevant question is whether the *ranking* of occupations changes materially. Figure C.1

reports pairwise Pearson and Spearman rank correlations of occupation-level expertise R_o across the five values of ρ .

All pairwise correlations are at least 0.88 (Pearson) and 0.94 (Spearman). Adjacent values, such as $\rho = 1$ and $\rho = 2$, correlate at about 0.99, and the lowest correlations are those between $\rho = 1$ and $\rho = 10$. The ranking of occupations is therefore largely preserved.

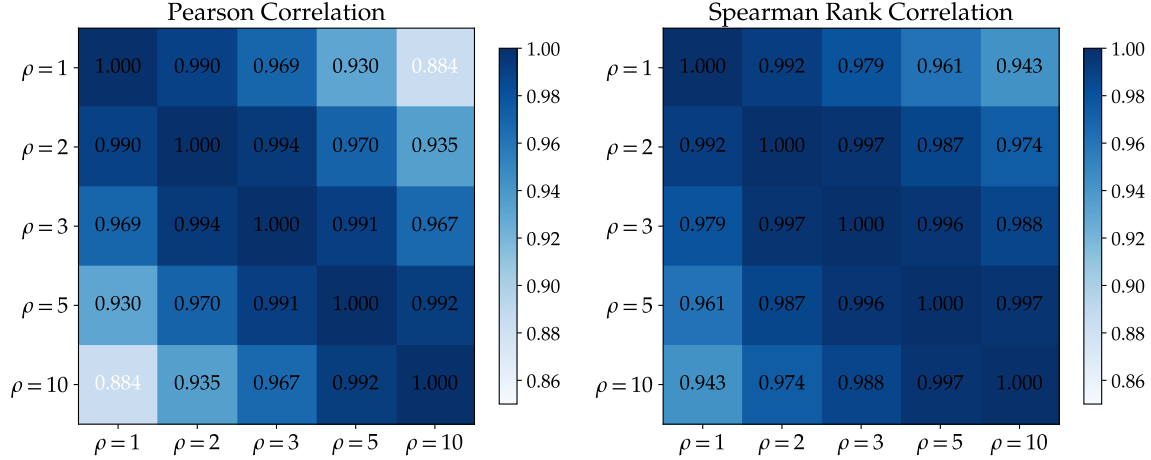


Figure C.1: Cross- ρ Correlations of Occupation-Level Expertise R_o

Notes: Each cell reports the pairwise correlation of occupation-level expertise requirements R_o computed under two different values of ρ in Equation (1). Left panel: Pearson correlation. Right panel: Spearman rank correlation. The sample consists of 923 occupations with non-missing task-level expertise and importance weights.

C.3.2 Potential Supply Shift (PSS)

Figure C.2 plots PSS against the baseline expertise percentile for each value of ρ . The pattern documented in the main text (Figure 3a) is present for all aggregation rules: PSS rises with baseline expertise over most of the distribution, peaks between roughly the 80th and 90th percentiles, and falls sharply in the top decile.

The level of PSS changes little with ρ (its mean falls from 0.117 at $\rho = 1$ to 0.114 at $\rho = 10$), because higher ρ raises both R_o and $\tilde{R}_o$ and the expertise distribution $F(\cdot)$ is rebuilt on the same scale. The shape is stable across aggregation rules.

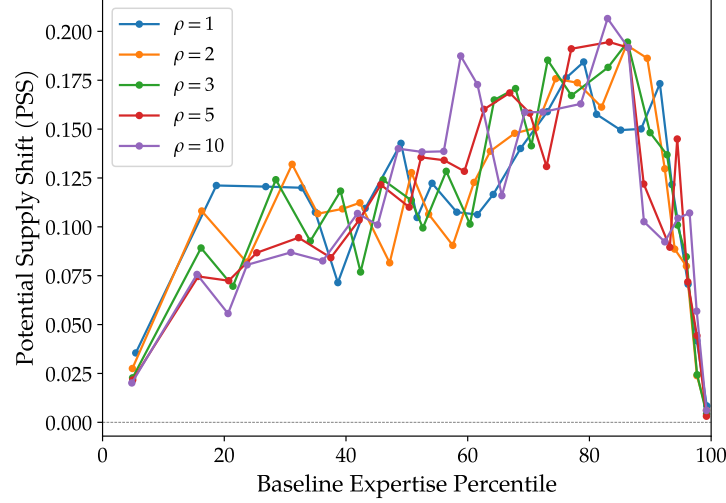


Figure C.2: PSS vs. Expertise Percentile Under Different ρ

Notes: Each line plots bin averages of the Potential Supply Shift (PSS) against occupation-level expertise percentiles, for a given value of ρ in the power-mean aggregator (1). PSS measures the change in the share of workers whose expertise exceeds the occupation's entry barrier. Occupations are grouped into 25 bins with equal numbers of occupations.

C.3.3 General-Equilibrium Wage Effects

Figure C.3 presents the distributional pattern of supply-channel wage changes across the pre-AI wage distribution (analogous to Figure 5 in the main text). The pattern is the same under all five aggregation rules: wages rise in low-wage occupations as workers leave for newly accessible, better-paid ones, and wage declines grow through the upper half of the distribution. The decline in the top bin, which contains the highest-expertise occupations, becomes smaller as ρ rises, because the power mean pushes those occupations further into the thin tail of F where few workers become newly eligible.

Table C.4 reports the effects on inequality across workers at our baseline of $\sigma = 4$, recalibrating τ for each value of ρ (it rises from 0.83 to 1.48). The results are largely consistent across values of ρ , except that the supply channel weakens as ρ rises. It lowers the variance of log wages by 8.3 percent at $\rho = 1$, 6.8 percent at $\rho = 2$, 4.8 percent at $\rho = 3$ and 2.3 percent at $\rho = 5$, and has essentially no effect at $\rho = 10$. The productivity channel raises the variance by 25 to 30 percent, and with both channels it rises by 13 to 24 percent.

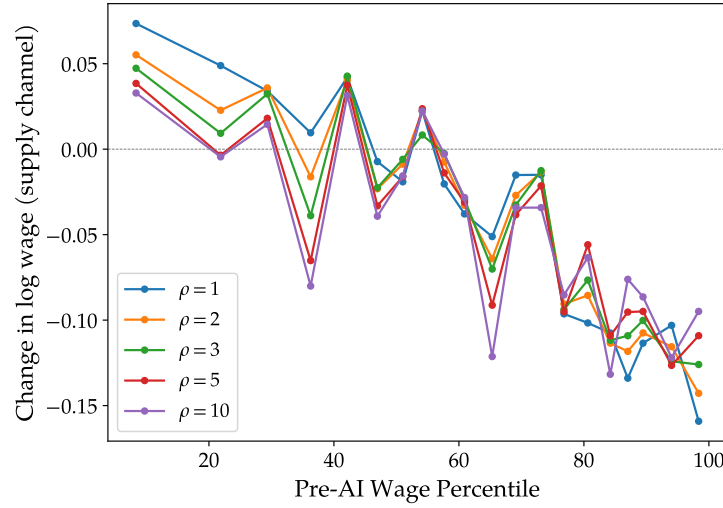


Figure C.3: Supply-Channel Wage Changes by Pre-AI Wage Percentile Under Different ρ

Notes: Each line plots employment-weighted bin averages of the change in log occupational wages (general equilibrium, supply channel only) against the occupation's percentile in the baseline wage distribution, for a given value of ρ . This figure is the multi- ρ analog of Figure 5. Occupations are grouped into 20 bins with equal numbers of occupations, so the bins differ from those of the main figure.

Table C.4: Inequality Effects by Power-Mean Exponent ρ

	$\rho = 1$	$\rho = 2$	$\rho = 3$	$\rho = 5$	$\rho = 10$
<i>Calibrated parameters</i>					
τ	0.833	1.005	1.134	1.313	1.477
Baseline $\text{Var}(\log w)$	0.164	0.157	0.153	0.150	0.149
<i>Percent change in $\text{Var}(\log w)$</i>					
Supply only	-8.3	-6.8	-4.8	-2.3	0.0
Productivity only	+24.9	+27.2	+28.3	+29.3	+30.3
Both channels	+13.1	+16.0	+18.6	+21.3	+23.9

Notes: Each column reports general-equilibrium results at $\sigma = 4$ under a different power-mean exponent ρ in Equation (1). τ is the logit taste parameter calibrated to match observed employment shares; the baseline variance differs across columns because the model's baseline employment, which weights the statistic, depends on the calibrated τ . "Supply only" isolates the wage effects of reduced entry barriers. "Productivity only" isolates the wage effects of higher occupational productivity. "Both channels" combines both mechanisms. All percentage changes are relative to the pre-AI baseline.

C.4 An Education-Based Expertise Distribution

The baseline analysis infers the distribution of worker expertise $F(\cdot)$ from occupation-level training requirements, weighting each occupation's requirement by its employment. Realized occupational training is an equilibrium outcome, not a primitive: it reflects preferences, licensing, and labor demand as well as the underlying distribution of ability, and it is constructed from the same requirements that define the entry barriers. This appendix replaces the occupation-based distribution with one built from the population-wide distribution of educational attainment, which is independent of the occupational requirements, and reports the counterfactuals under that distribution. The exercise keeps everything else fixed: the same R_o and $\tilde{R}_o$, the same productivity gains, $\sigma = 4$, and $\tau = 0.833$, calibrated on the baseline distribution and held fixed so that the comparison isolates the change in F .

C.4.1 Construction of the CPS Education-Based Distribution

We use Table 2 of the Census Bureau's *Educational Attainment in the United States: 2024*, drawn from the CPS Annual Social and Economic Supplement (ASEC). This table reports the number of individuals aged 25 and over by highest educational attainment, which we map to months of post-high-school education as follows:

This yields a distribution with a mean of 30.0 months of education beyond high school. We construct the CDF by linearly interpolating between the discrete mass points, which gives a median of about 16 months; for each occupation o with training requirement R_o (in months), the share of the population qualified is $S_o = 1 - F_{\text{CPS}}(R_o)$, and likewise for $\tilde{R}_o$.

C.4.2 Comparison of Expertise Distributions

Figure C.4 compares the baseline (occupation-based) and CPS (education-based) expertise CDFs. The two distributions differ in an important way. The baseline distribution has most of its mass where most occupations' entry barriers lie: 60 percent of workers have an expertise level between 1 and 12 months, the range that contains about two thirds of occupations. The CPS distribution, interpolated between degree-completion points, places

Table C.5: Mapping Educational Attainment to Months of Post-High-School Education

Category	Pop. (thousands)	Months post-HS	Share (HS+)
High school graduate (incl. GED)	64,010	0	0.305
Some college, no degree	32,170	12	0.153
Associate's degree	25,060	24	0.119
Bachelor's degree	54,560	48	0.260
Master's degree	26,010	72	0.124
Professional degree (JD, MD)	3,211	84	0.015
Doctoral degree (PhD)	5,113	108	0.024
Total (HS+)	210,134		1.000

Source: U.S. Census Bureau, CPS ASEC 2024. Individuals without a high school diploma (19,643 thousand, 8.5 percent of the population aged 25 and over) are excluded, so the distribution measures education beyond high school. Shares are renormalized to sum to one over the HS+ population.

30 percent of workers at zero months and spreads the rest thinly over several years of schooling, so that only 14 percent of workers fall between 1 and 12 months; its scale is years of formal schooling rather than task-specific training time. The two are nonetheless positively related at the occupation level: the Pearson correlation between $F_{\text{baseline}}(R_o)$ and $F_{\text{CPS}}(R_o)$ across the 883 occupations is 0.76 (Figure C.4b).

The difference in shape has a direct consequence for the size of the supply shock. Because the CPS distribution is thin in the range where barriers fall, a given reduction from R_o to $\tilde{R}_o$ makes far fewer workers newly eligible. The mean PSS is 0.038 under the CPS distribution against 0.117 under the baseline, and for 31 percent of occupations it is below 0.01 in absolute value. The education-based distribution therefore understates the supply shift relative to the baseline, for reasons that have to do with how its mass is distributed and with its unit rather than with the economics. The results below should be read with this in mind.

C.4.3 Counterfactual Results

Table C.6 reports the inequality statistics and the decomposition of Appendix C.1 under the CPS distribution, at our baseline of $\sigma = 4$. The signs of both channels are the same as in the baseline. The supply channel lowers the variance of log wages across workers by

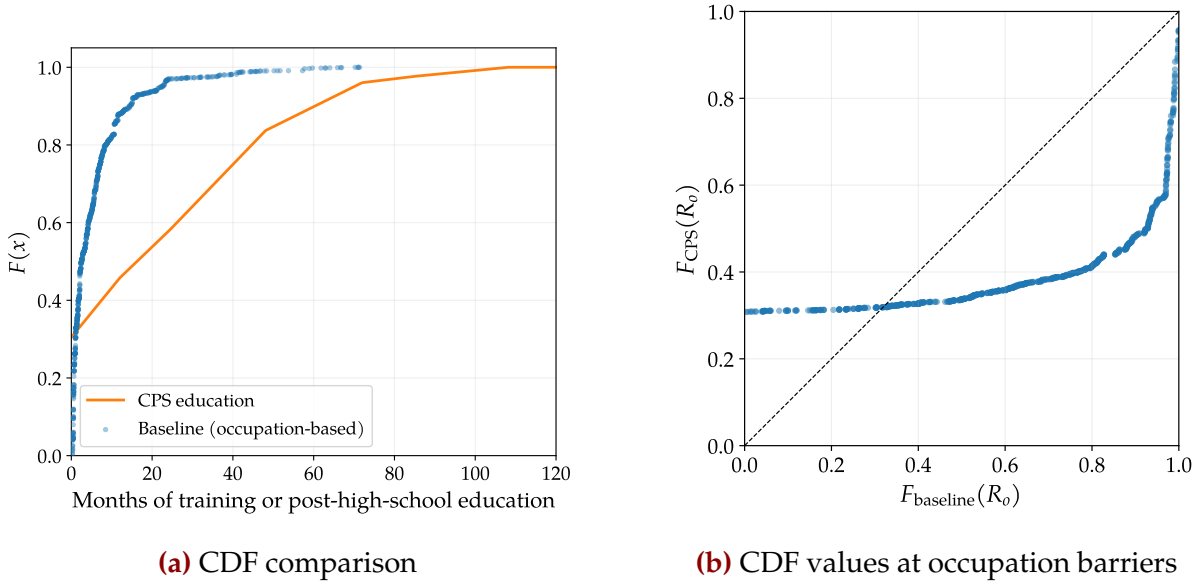


Figure C.4: Baseline and CPS Education-Based Expertise Distributions

Notes: Panel (a) plots both CDFs over months of training or post-high-school education. Panel (b) plots $F_{baseline}(R_o)$ against $F_{CPS}(R_o)$, evaluated at the training requirement of each of the 883 occupations.

7.4 percent and compresses the upper half of the distribution ($p_{90} - p_{50}$ falls from 0.631 to 0.582); the productivity channel raises the variance by 14.3 percent; and the two together raise it by 4.5 percent. The composition of the supply effect differs from the baseline: with the smaller supply shock, the wage-structure effect is -7.1 percent of baseline variance and the composition effect is close to zero (-0.3), so the total across workers is essentially the wage-structure effect.³⁷

Table C.7 compares the two distributions across values of σ . The race works the same way under both. The supply channel lowers inequality across workers at every σ (except at $\sigma = 8$ in the baseline), the productivity channel raises it at every σ , and with both channels inequality falls at $\sigma = 1.5$ and $\sigma = 2$ and rises at all higher values; the threshold is about $\sigma = 2.9$ under the CPS distribution, against 2.7 in the baseline. The productivity channel is smaller under the CPS distribution at every σ , and the supply channel's effect depends less on σ than in the baseline (from -14 to -4 percent, against -21 to $+1$ percent), because its wage-structure effect is less than half as large at every σ ; and since its

³⁷The model's baseline employment under the CPS distribution is essentially uncorrelated with observed employment (correlation -0.02), because the CPS distribution gives many occupations nearly identical eligible pools.

Table C.6: Wage Inequality Under the CPS Education-Based Expertise Distribution

<i>Panel A: Inequality statistics</i>				
Scenario	$\text{Var}(\log w)$	$p90 - p10$	$p50 - p10$	$p90 - p50$
Baseline	0.247	1.223	0.592	0.631
Supply channel	0.229	1.168	0.586	0.582
Productivity channel	0.282	1.371	0.698	0.673
Both channels	0.258	1.292	0.665	0.627
<i>Panel B: Decomposition of the change in $\text{Var}(\log w)$, percent of baseline variance</i>				
	Total	Wage structure	Composition	
Supply channel	-7.4	-7.1	-0.3	
Productivity channel	+14.3	+16.5	-2.2	
Both channels	+4.5	+8.1	-3.7	

Notes: Same construction as Tables 3 and C.1, with the expertise distribution F replaced by the CPS education-based distribution of Table C.5; $\sigma = 4$ and $\tau = 0.833$ as in the baseline. Panel A reports employment-weighted moments of log occupational wages, each scenario weighted by its own employment. Panel B decomposes the general-equilibrium change in the variance into a wage-structure effect (new wages at baseline employment) and a composition effect (reallocation of workers).

composition effect is close to zero, it stays negative even at $\sigma = 8$.

Table C.7: Change in Inequality Across Workers by Elasticity of Substitution: Baseline and CPS Expertise Distributions

σ	1.5	2	3	4	5	6	8
<i>Panel A: Baseline expertise distribution (Table 4)</i>							
Supply channel	-21.5	-18.2	-12.6	-8.3	-5.1	-2.7	+0.9
Productivity channel	+7.8	+13.2	+20.3	+24.9	+28.0	+30.4	+33.6
Both channels	-17.1	-9.4	+3.5	+13.1	+20.2	+25.7	+33.5
<i>Panel B: CPS education-based expertise distribution</i>							
Supply channel	-13.5	-11.6	-9.0	-7.4	-6.3	-5.5	-4.3
Productivity channel	+4.3	+7.4	+11.6	+14.3	+16.1	+17.5	+19.4
Both channels	-10.2	-5.7	+0.5	+4.5	+7.3	+9.3	+12.2

Notes: Percent change in the employment-weighted variance of log occupational wages across workers between the pre-AI baseline and each general-equilibrium counterfactual. Panel A repeats Table 4. Panel B replaces the expertise distribution F with the CPS education-based distribution of Table C.5. In both panels $\tau = 0.833$ and $\{B_o\}$ is recalibrated for each σ .

C.4.4 Discussion

Two conclusions follow. First, the direction of each channel, and how the race depends on σ , do not depend on how the expertise distribution is measured: under both distributions, lower entry barriers compress the occupational wage structure, productivity gains widen it, and the net effect across workers changes sign at about the same σ . Second, the magnitude and composition of the supply channel do depend on it. The education-based distribution gives a supply shock about a third the size of the baseline, for the reason described above, and, at every σ , a wage-structure effect less than half as large and a composition effect close to zero rather than large and positive. We regard the baseline distribution as the better proxy for the purpose of measuring PSS, because it is continuous in the same units as the task-level requirements, but we do not have a measure of worker expertise that is both independent of occupational requirements and fine enough to locate workers relative to the thresholds. Building such a measure from worker-level data on education and experience is the natural next step and is beyond the scope of this paper.

D Dynamic Model

D.1 Additional Details

This appendix collects details of the dynamic model of Section 6. The prompts for the retraining distances are in Appendix B.1.5. For 23 percent of occupation pairs the LLM reports a longer retraining time with GenAI; because access to an assistant cannot make retraining slower, we treat these increases as noise and set $d_{so}^{\text{AI}} = d_{so}$ for these pairs (without this, the long-run decline in the variance of log wages is 11.5 rather than 9.8 percent). This differs from the static model, where we keep requirements that rise with GenAI: there, a requirement can rise when automation removes an occupation’s easier tasks, whereas here GenAI enters only as an aid to learning. Table D.1 summarizes the calibration. As in the static model, the common term $\Gamma = Y^{1/\sigma}$ in B_o rises with aggregate output in the post-AI steady state; this raises all post-AI wages by 1.7 percent and affects neither the allocation of workers nor inequality. Because the present value of a permanent flow amenity a_o is amplified by $1/(1 - \beta)$ through the Bellman recursion, the contraction mapping that recovers amenities applies a $(1 - \beta)$ correction when updating them. Figure D.1 plots the LLM-based retraining distances against the O*NET skill-shortfall benchmark.

Table D.1: Dynamic Model: Calibrated Parameters

Parameter	Role	Value	Source / Target
β	Discount factor	0.95	Standard
σ	CES elasticity	4.0	Same as static model
τ	Choice dispersion	0.463	Set (not identified); see Appendix D.2
$\{a_o\}, \{B_o\}$	Amenities, productivity	inverted	Observed employment and wages
κ^{out}	Origin stickiness (J)	calibrated	Origin outflow rates $\{m_s^{\text{data}}\}$
κ^{in}	Destination barriers (J)	calibrated	Destination shares $\{f_o^{\text{data}}\}$

Notes: τ comes from an earlier version of the calibration and is held fixed; it is not identified by the calibration moments (Appendix D.2). The $2J$ switching-cost parameters are calibrated to the flow moments of Schubert et al. (2024).

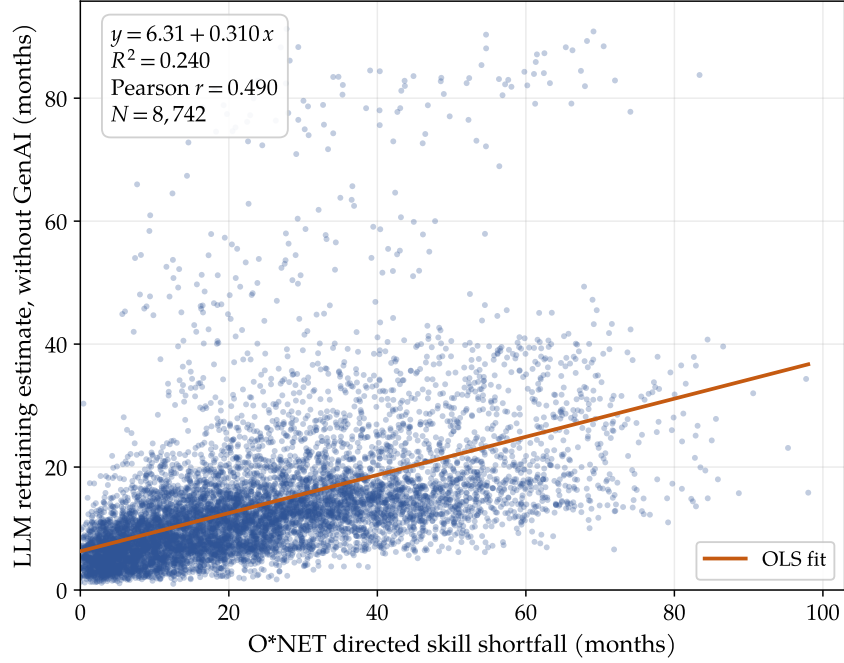


Figure D.1: Validation: LLM Retraining Estimates vs. O*NET Skill Shortfall

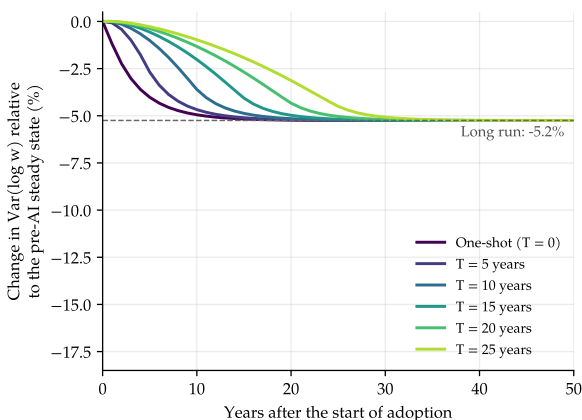
Notes: Each point is an ordered pair of 3-digit SOC occupations ($N = 8,742$ off-diagonal pairs). The x-axis is the O*NET directed skill shortfall d_{so}^{O*NET} (Section 6.2), rescaled to a mean of two years, and the y-axis the LLM-based retraining estimate, both in months. The line is an OLS fit. Pearson correlation 0.49; Spearman rank correlation 0.65.

D.2 Sensitivity to the Dispersion of Taste Shocks

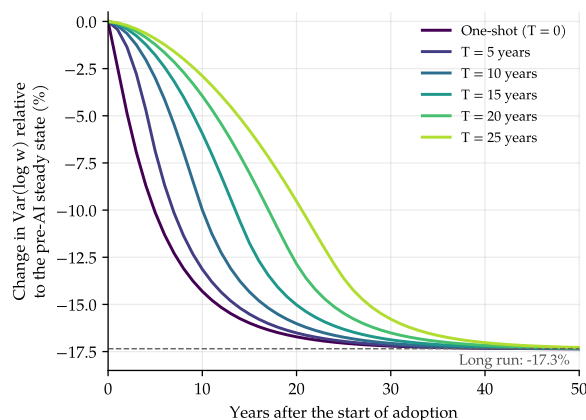
The calibration of Section 6.3 does not identify τ : because the pre-AI steady state reproduces observed wages and employment by construction, observed mobility pins down switching costs only relative to τ . This appendix reports results for a lower value, $\tau = 0.25$, and a higher one, $\tau = 0.83$, the value calibrated in the static model, recalibrating the switching-cost parameters ($\kappa^{out}, \kappa^{in}$) in each case. The model fits the flow moments equally well under all three values of τ (correlations of 0.98 for outflow rates and 0.999 for destination shares).

Figure D.2 shows the transition paths. The long-run decline in the variance of log wages is 5.2 percent with $\tau = 0.25$ and 17.3 percent with $\tau = 0.83$, against 9.8 percent in the baseline. Because observed mobility pins down switching costs relative to τ , a higher τ makes workers respond less to wages: falling wages in high-wage occupations

then deter entry less, so more workers move and scarcity premia erode more. The timing barely changes. After a decade, the share of the long-run decline that has been realized is 94 percent with $\tau = 0.25$ and 82 percent with $\tau = 0.83$ if retraining distances fall at once (89 percent in the baseline), 69 and 58 percent if adoption takes 10 years (63 percent), and 19 and 17 percent if it takes 25 years (18 percent).



(a) $\tau = 0.25$



(b) $\tau = 0.83$

Figure D.2: Transition Paths Under Alternative Values of τ

Notes: Each panel repeats panel (b) of Figure 8 for a different dispersion of taste shocks τ (baseline $\tau = 0.46$), with the switching-cost parameters recalibrated to the flow moments: the percentage change in the variance of log wages relative to the pre-AI steady state, weighted by employment in each year, for adoption periods T from 0 (one-shot) to 25 years. The dashed line marks the long-run change. Both panels use the same vertical axis.